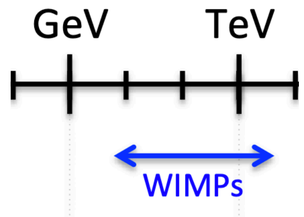
Searching for Ultraheavy and Ultralight Dark Matter

Kevin Zhou

SLAC Seminar— March 18, 2022

The Dark Matter Landscape



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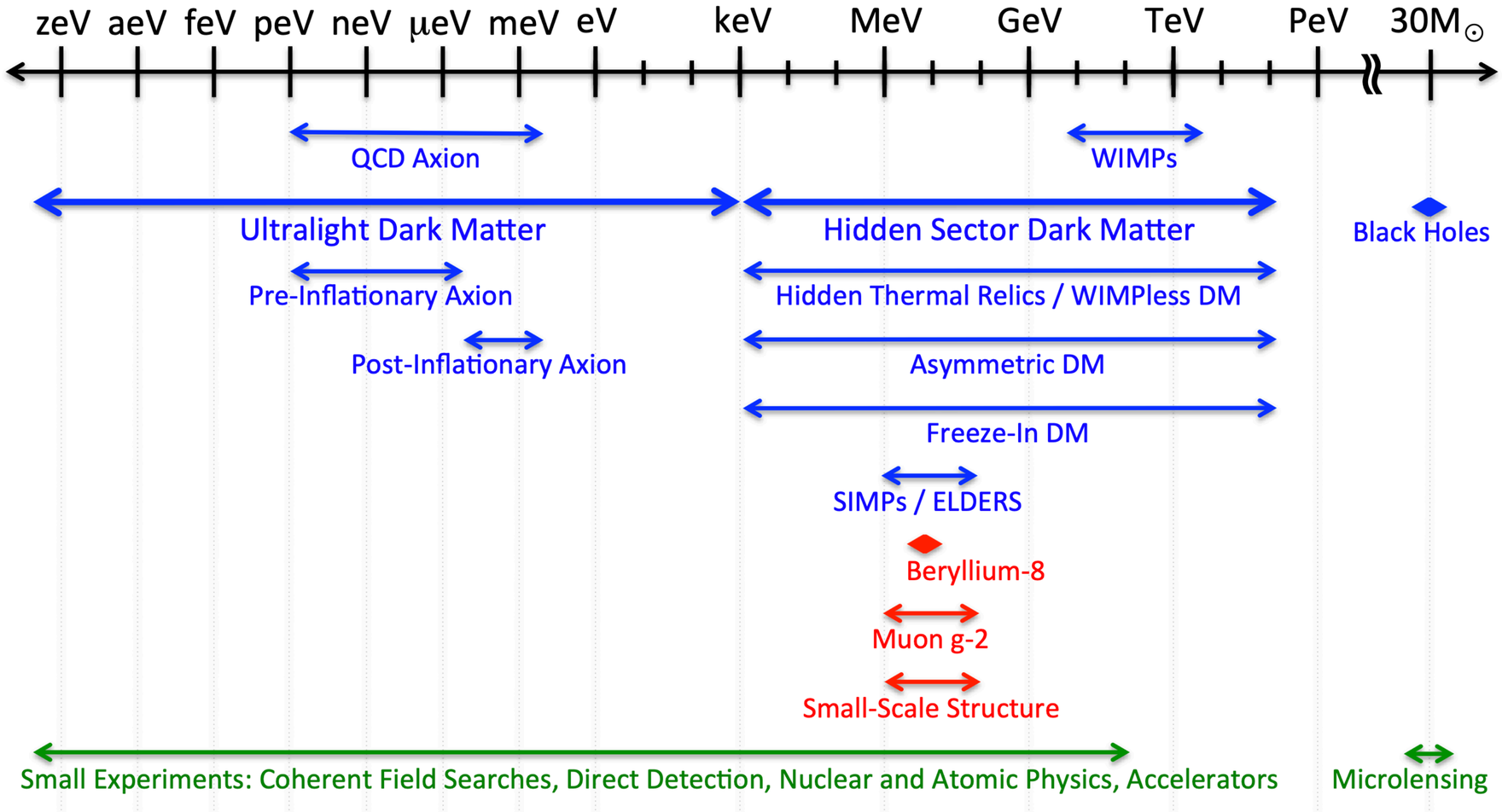
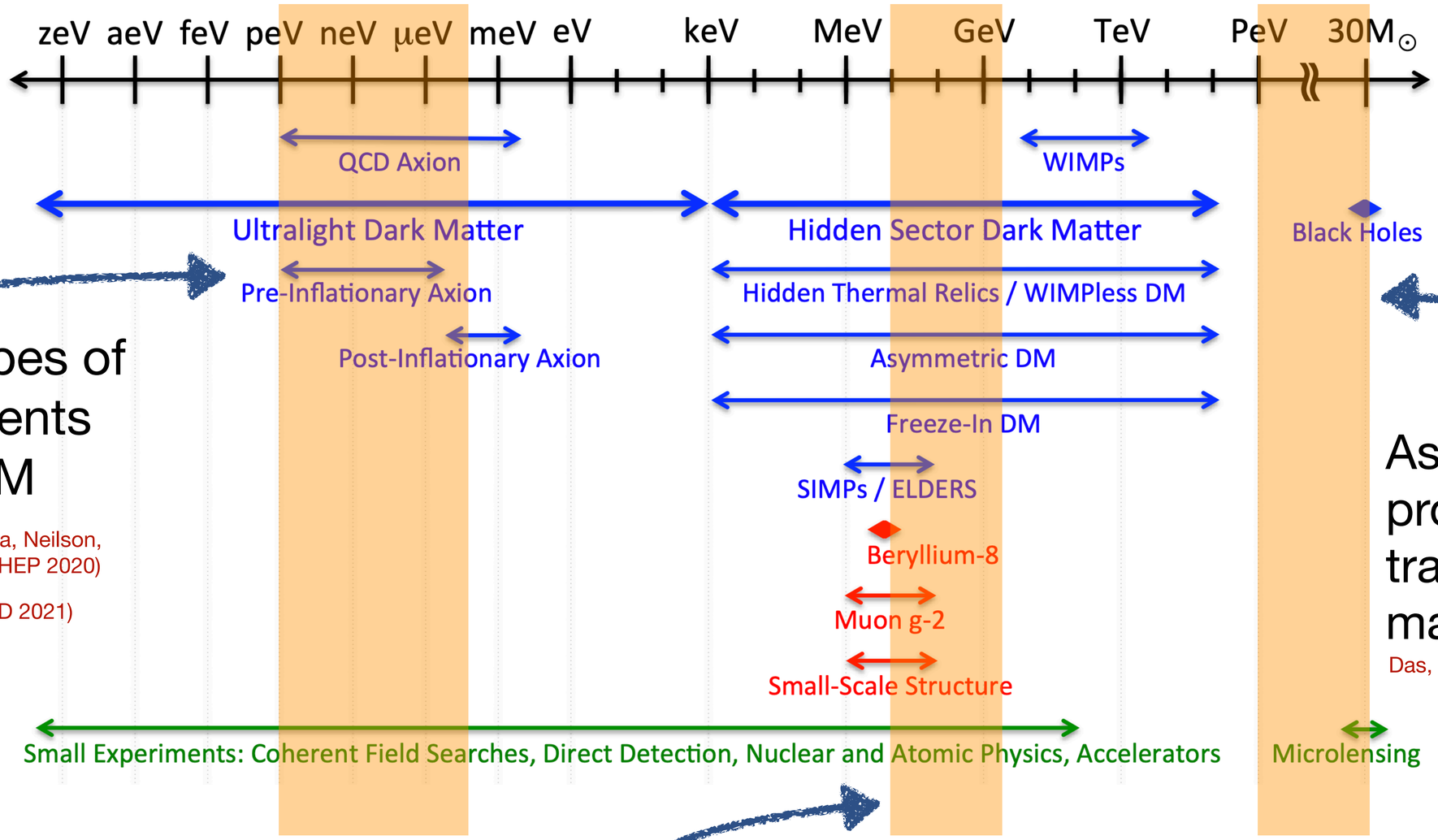


Cosmic Visions 1707.04591

The Dark Matter Landscape



Tabletop probes of effective currents from axion DM

Berlin, D'Agnolo, Ellis, Nantista, Neilson, Schuster, Tantawi, Toro, KZ (JHEP 2020)

Berlin, D'Agnolo, Ellis, KZ (PRD 2021)

Berlin, KZ (in prep)

Astrophysical probes of transients from macroscopic DM

Das, Ellis, Schuster, KZ (PRL 2022)

Vector meson decays to DM at LDMX

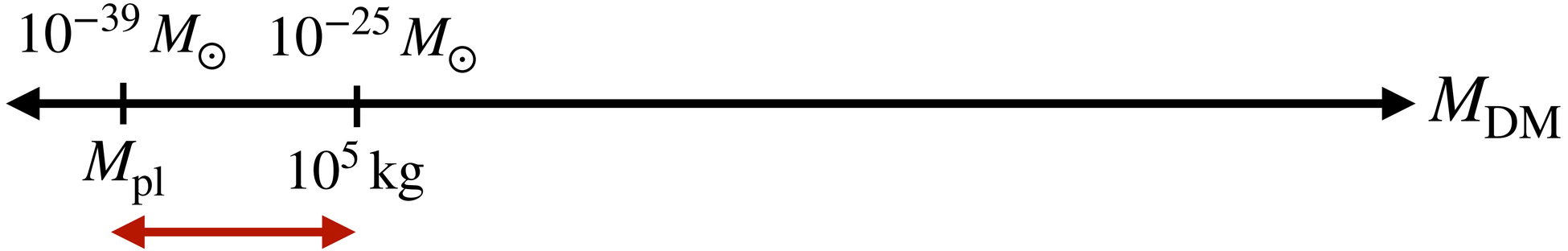
Schuster, Toro, KZ (PRD 2022)

Macroscopic Dark Matter

- Wide mass range available
- Forms in a variety of models
  - Direct fusion in asymmetric DM: “dark nucleosynthesis”
    1411.3739, 1707.02316
  - Collapse through cooling via dark $U(1)$
    1707.03829, 1812.07000
  - Phase transitions forming “nuggets” or solitons
    1810.04360, 2105.02840

Macroscopic DM Searches

Strong DM-SM interactions allowed, but still hard to detect because of rarity



"Low" masses probed by terrestrial searches:

WIMP/neutrino detectors
1803.08044, 1812.09325

Etched plastic
2012.13406

Ancient mica
2105.06473

Gravitational wave detectors
1807.03788

Meteors and cosmic rays
2008.01285

Seismic waves
1610.09680

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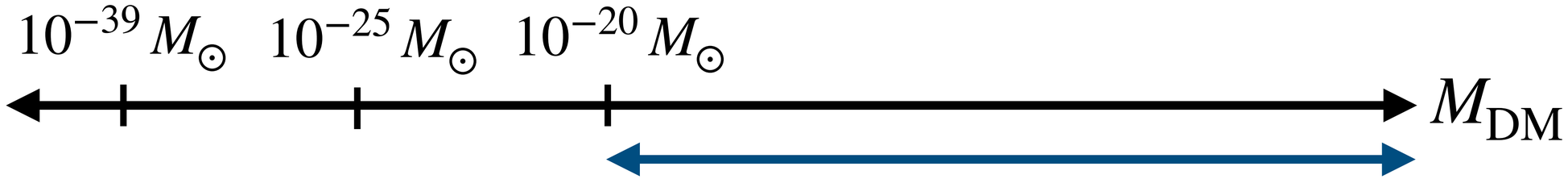

For higher masses, too rare to detect on Earth, $\Gamma \lesssim 1/(10^4\,\mathrm{yr})$

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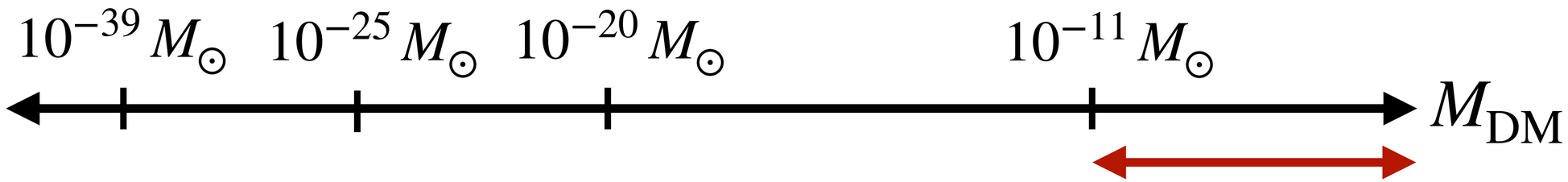

Highest masses excluded by gravitational lensing

1701.02151

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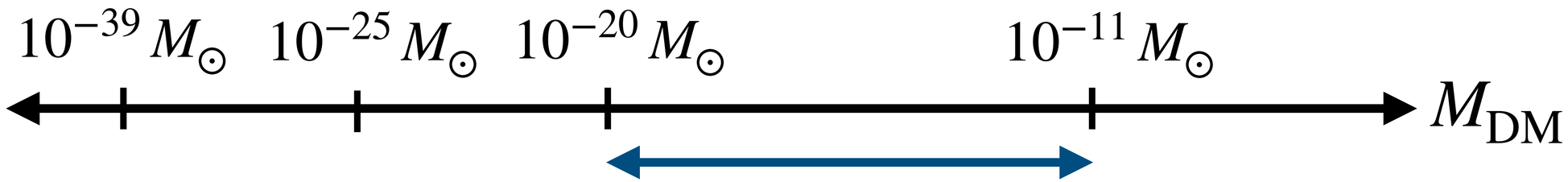
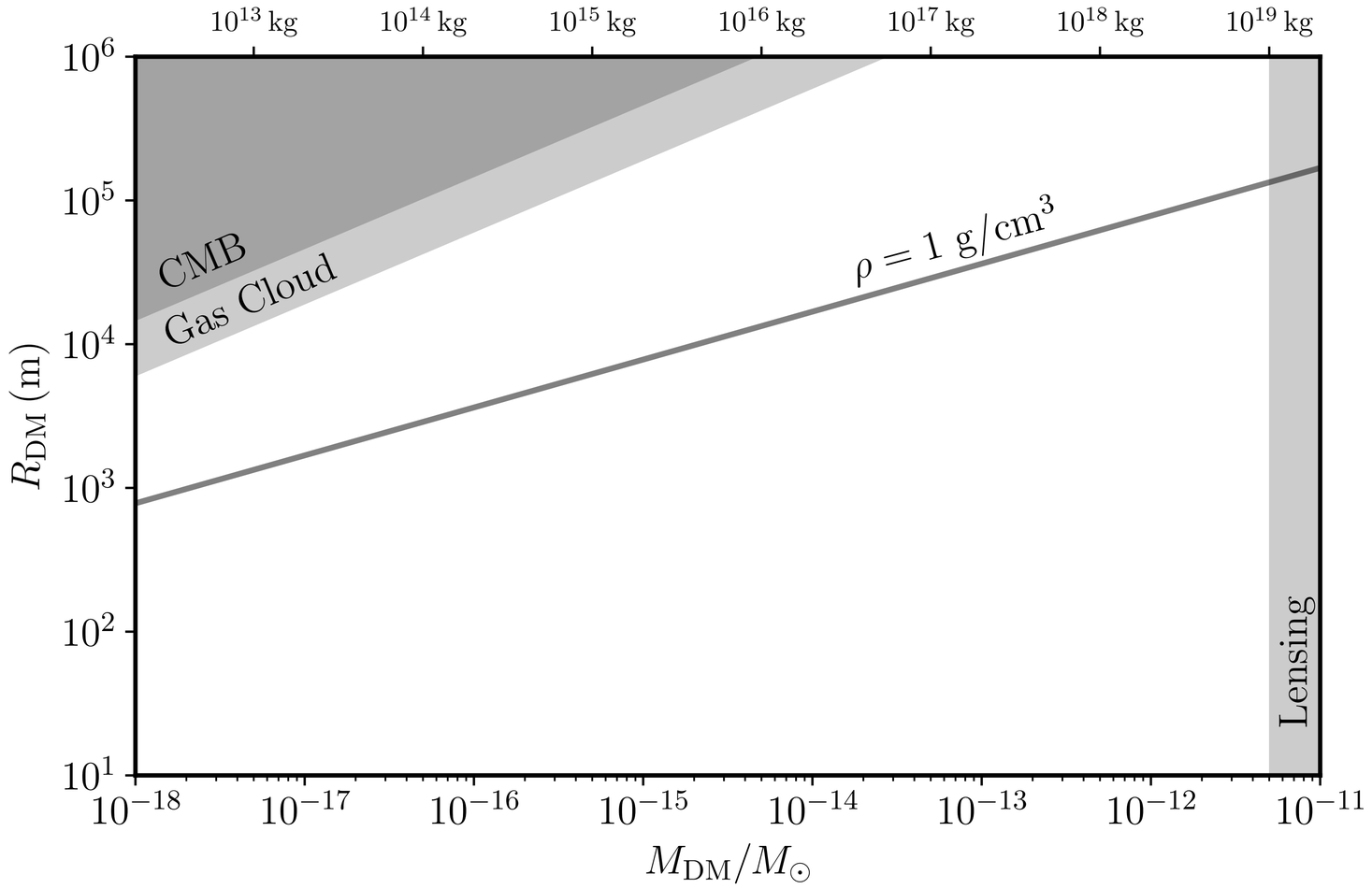
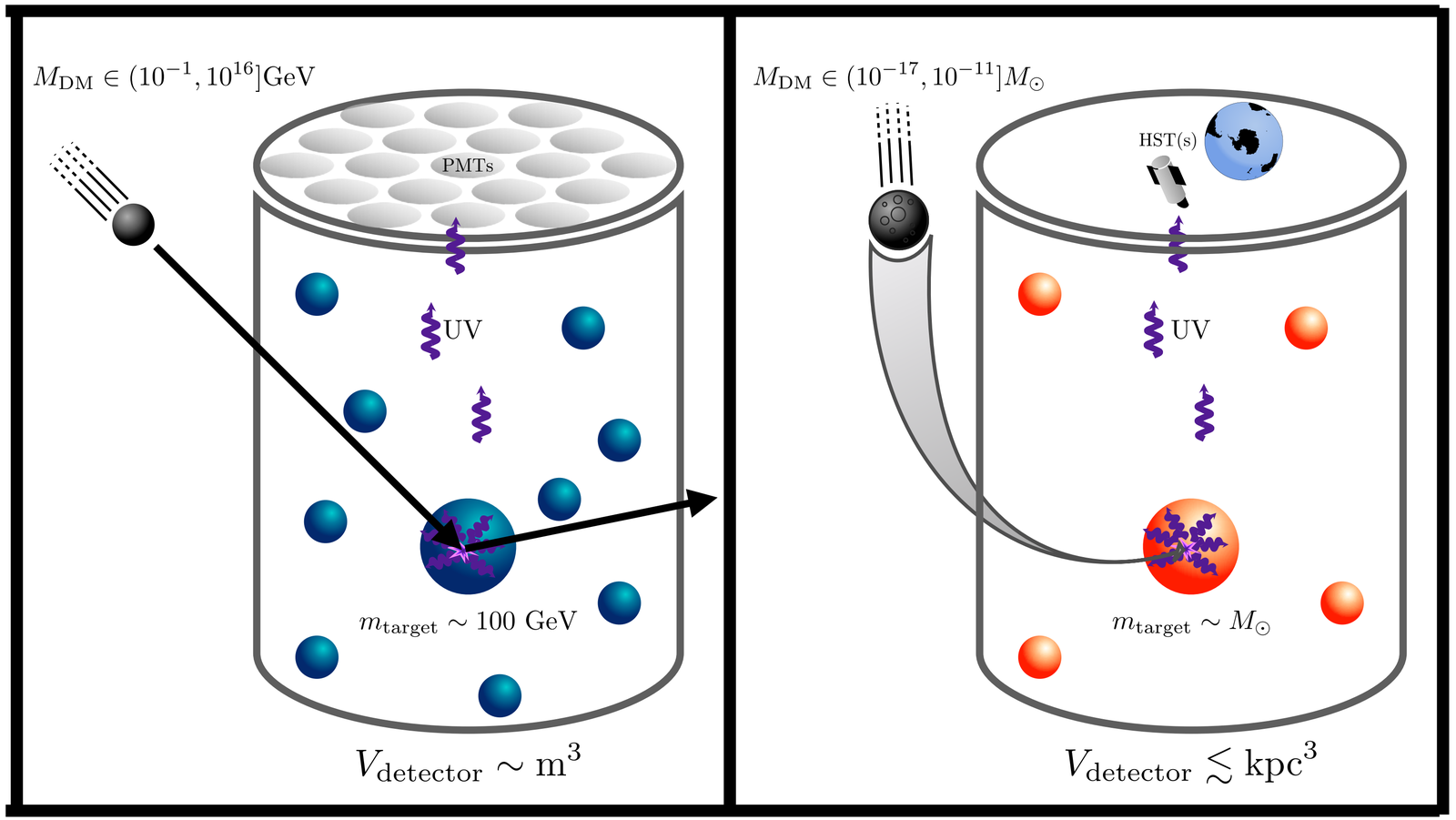


A relatively unconstrained “dark asteroid” range



Few existing constraints, near atomic density and higher, even for geometric cross section $\sigma = \pi R_{\rm DM}^2$

Probe by looking for stellar impacts: a dark matter direct detection experiment on astronomical scales!



Naive Estimates

Rate per star:

$$\Gamma \sim \frac{\rho_{\rm DM} v_{\rm DM}}{M_{\rm DM}} \pi R_\star^2 \left(1 + \frac{v_{\rm esc}^2}{v_{\rm DM}^2}\right) \sim \frac{1}{10^4\,{\rm yr}} \frac{10^{-15} M_\odot}{M_{\rm DM}}$$

Energy dissipated in star:

$$E_{\rm DM} \sim \frac{1}{2} M_{\rm DM} v_{\rm esc}^2 \sim 10^{34}\,{\rm erg}\,\frac{M_{\rm DM}}{10^{-15} M_\odot}$$

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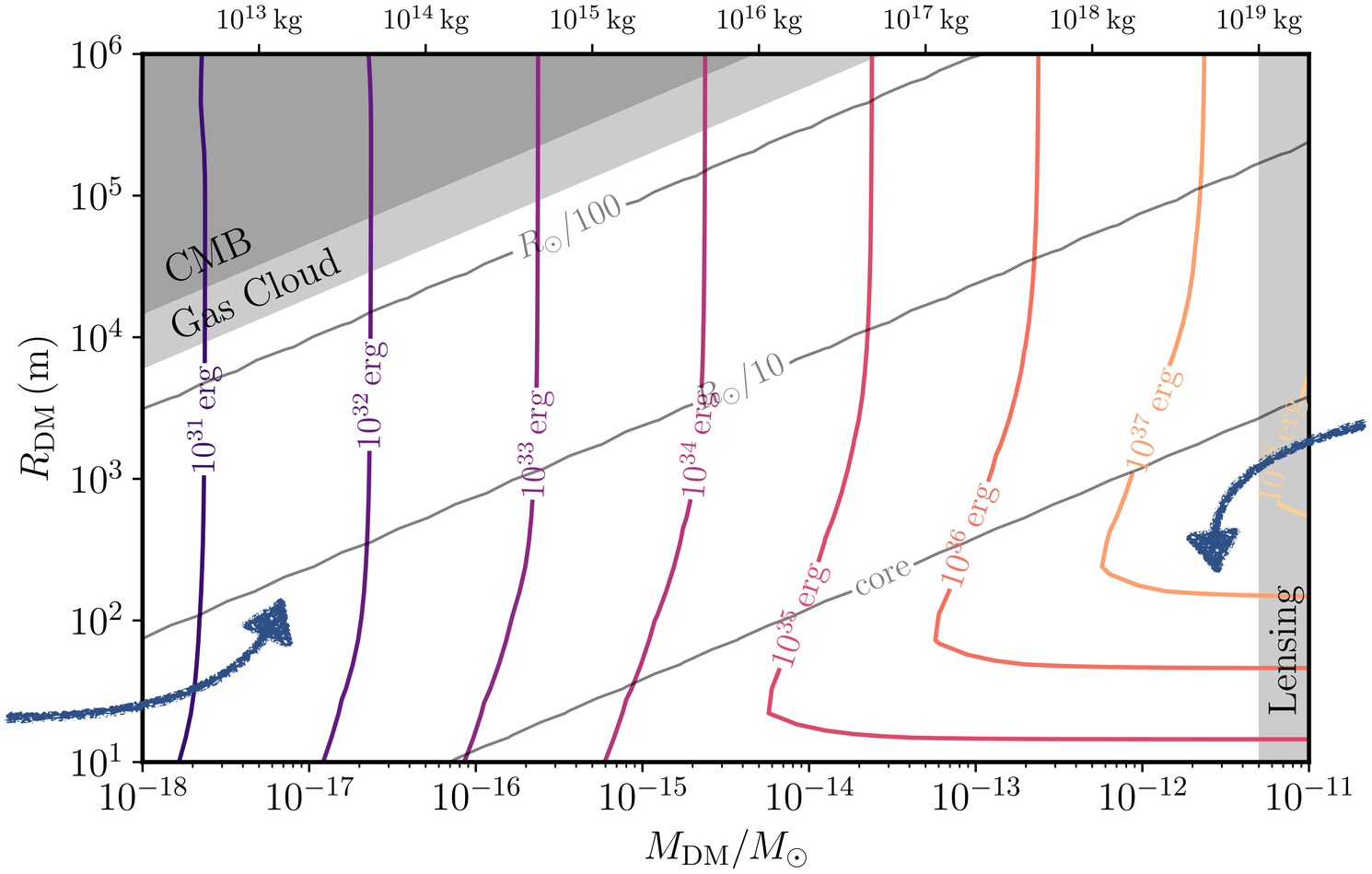
Average power $\Gamma E_{\rm DM}$ negligible, but $E_{\rm DM}$ high, potentially observable as strong transient



Penetration depth

Energy **dissipated** in star

Dark asteroid experiences gravitational force $M_{\rm DM}g$ and drag $F_d \sim \rho\sigma v^2$

Taking geometric cross section $\sigma = \pi R_{\rm DM}^2$, but qualitatively similar for lower σ

Rigidity and Opacity

Ordinary solids are very weakly bound because of several small numbers:

$$\frac{E_b}{mc^2} \sim \frac{\text{intermolecular bond energy}}{\text{nuclear mass}} \sim \frac{\alpha^2 m_e/10}{N m_p} \sim 10^{-10}$$

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Asteroids fall apart in stellar atmosphere because energy dissipated is much larger:

$$\frac{E_k}{mc^2} \sim \frac{v_{\text{esc}}^2}{c^2} \sim 10^{-6}$$

Not hard to have weakly interacting dark sectors with $E_b/mc^2 \gtrsim 10^{-6}$, where dark asteroid remains intact throughout whole impact

Rigidity and Opacity

Geometric cross section does not need unreasonable DM-SM interactions, since dark asteroids are large:

$$\sigma \lesssim \frac{1}{n_{\rm DM} R_{\rm DM}} \sim \frac{m_{\rm DM}}{\rho_{\rm DM} R_{\rm DM}} \sim (10^{-30}\,{\rm cm}^2)\,\frac{m_{\rm DM}}{0.1\,{\rm GeV}}\,\frac{1\,{\rm g/cm}^3}{\rho_{\rm DM}}\,\frac{1\,{\rm km}}{R_{\rm DM}}$$

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For light constituents, σ below direct detection bounds even if most DM is unbound

If DM-DM scattering also geometric, bounds on $\sigma/m_{\rm DM}$ weaker by orders of magnitude

How the Energy Escapes

Naively, energy deposited even at depth $R_\star/10$ takes **very** long time to get out

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But dark asteroid is traveling supersonically, at $\mathrm{Ma} = v_{\mathrm{esc}}/c_s \sim 100$, so energy dissipated forms shock waves, which efficiently propagate to surface!

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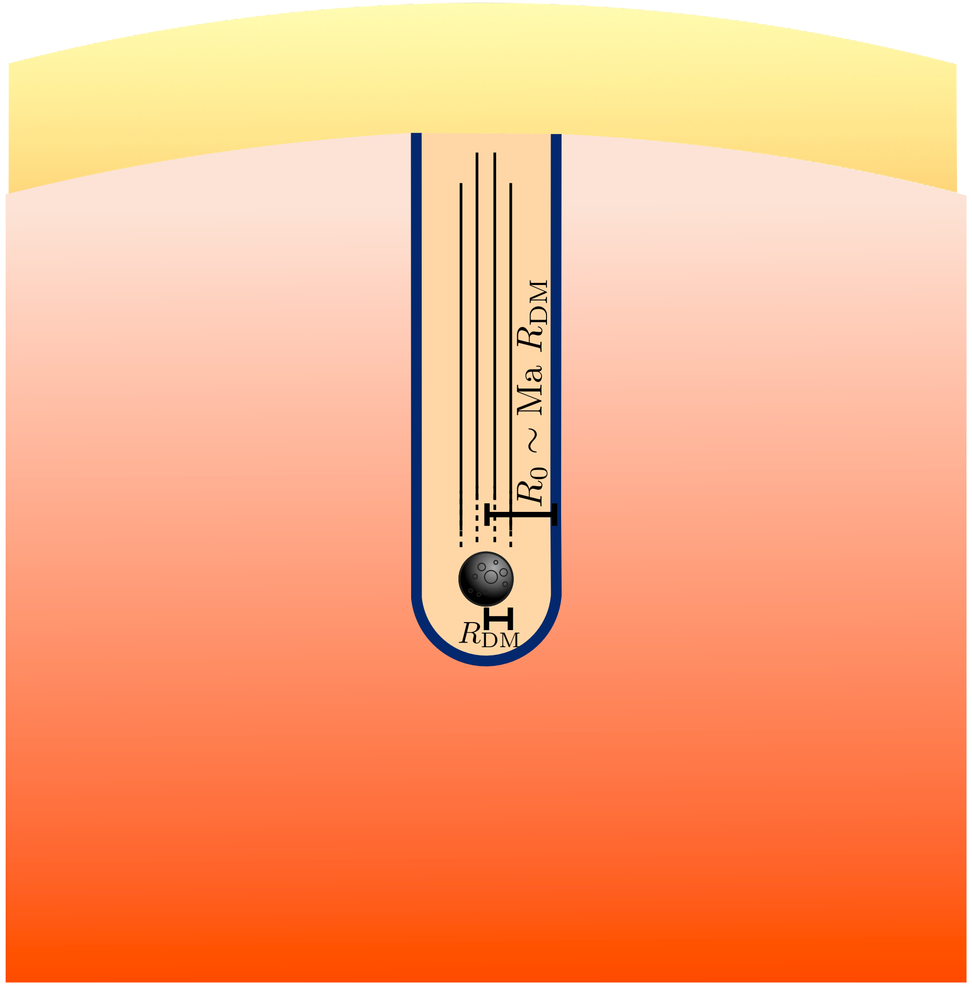
A complicated hydrodynamic problem, but solvable with controlled approximations known for decades

Whitham (1956), ReVelle (1976)

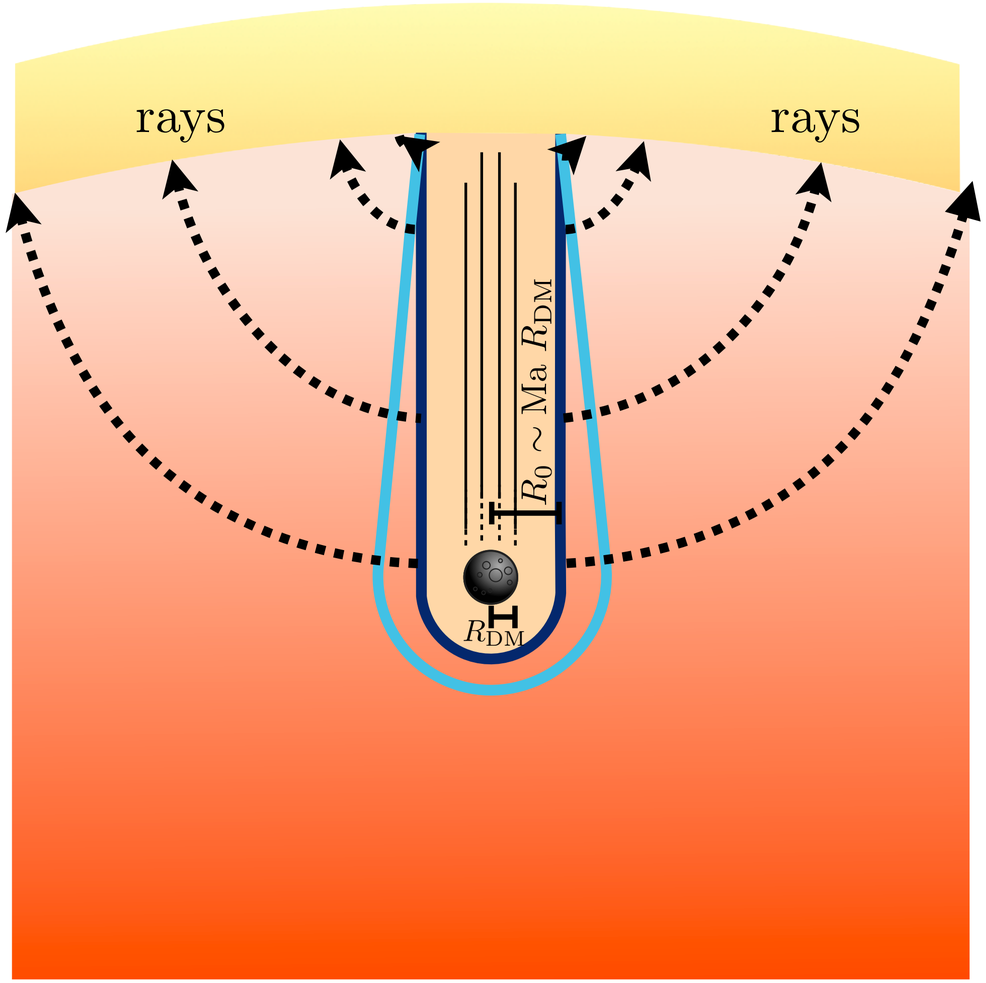
How the Energy Escapes



Near the dark asteroid: since $\mathrm{Ma} \gg 1$, can approximate with known cylindrical blast wave

Characteristic size $R_0 \sim \mathrm{Ma}\, R_{\mathrm{DM}}$

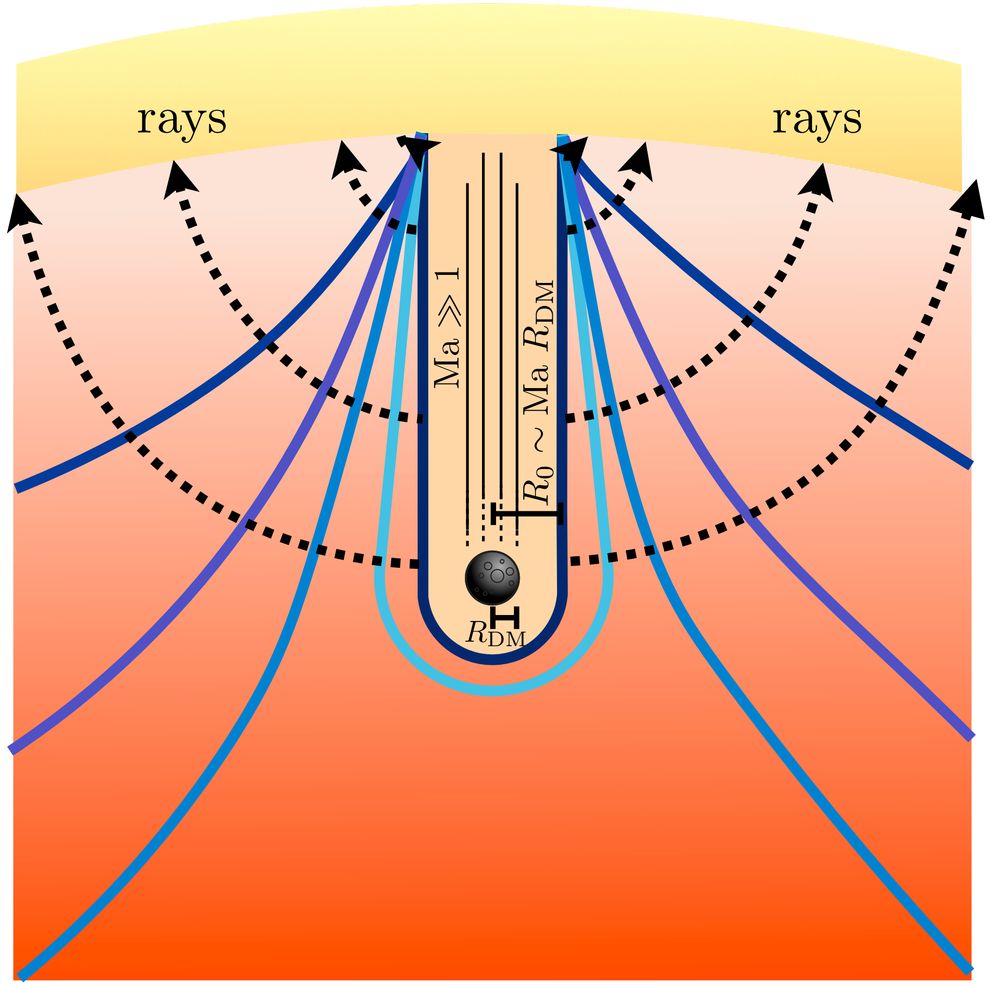
How the Energy Escapes



Beyond R_0, match onto a weak shock solution — essentially an acoustic wave with additional dissipation

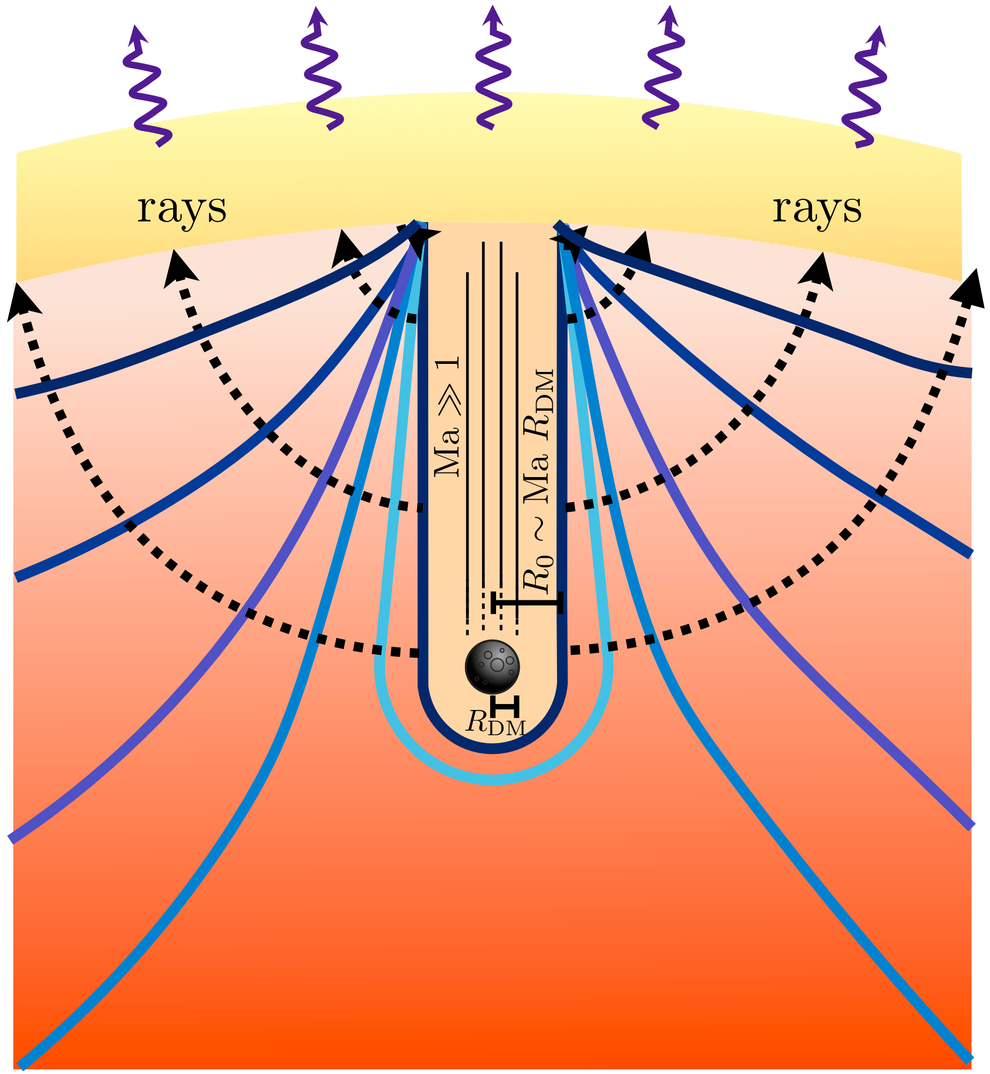
Rays refract radially outward

How the Energy Escapes

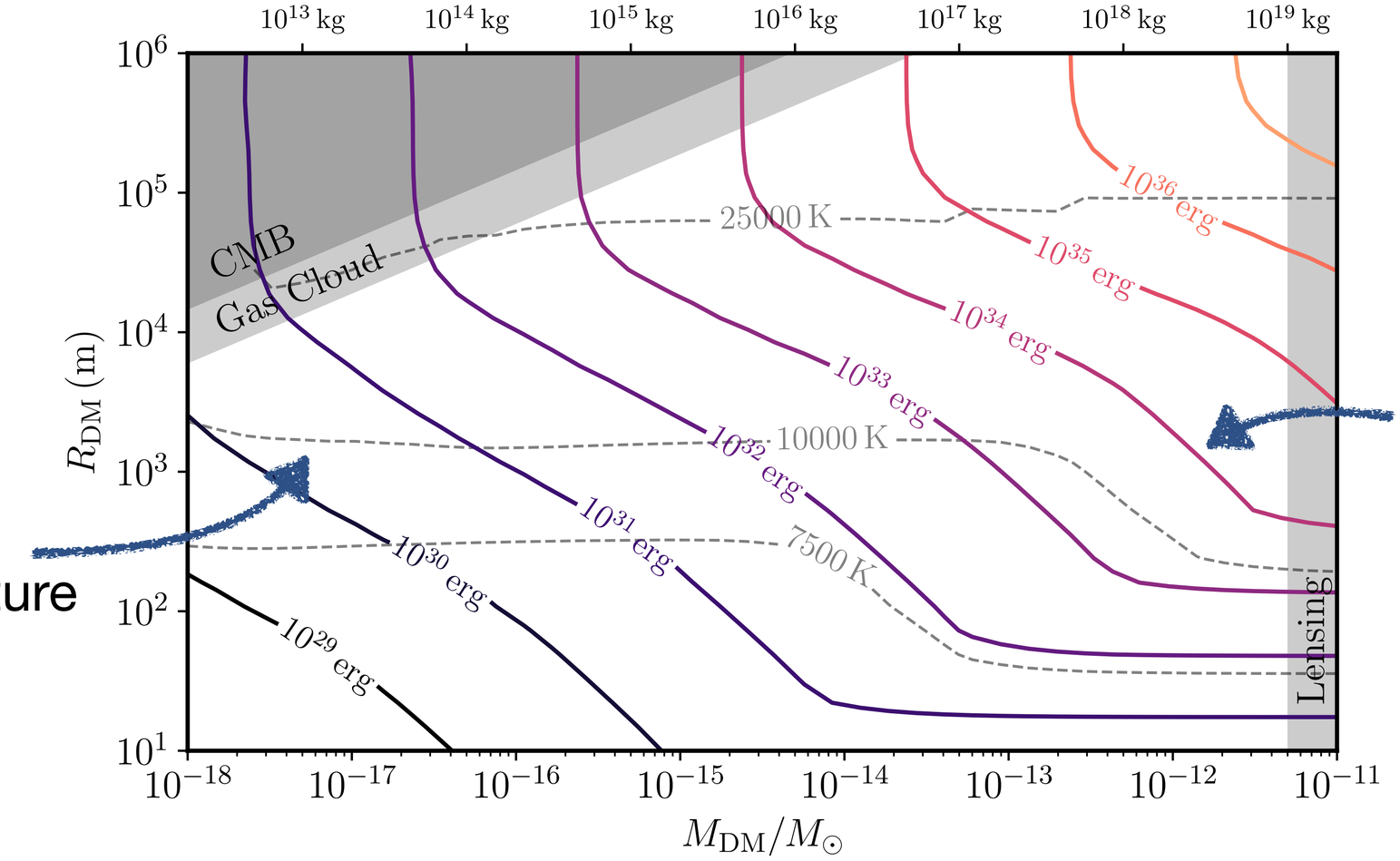


Weak shock waves lose energy, but grow in strength as they approach the stellar surface

How the Energy Escapes



Shock waves grow strong very close to the photosphere, heating a patch of the surface

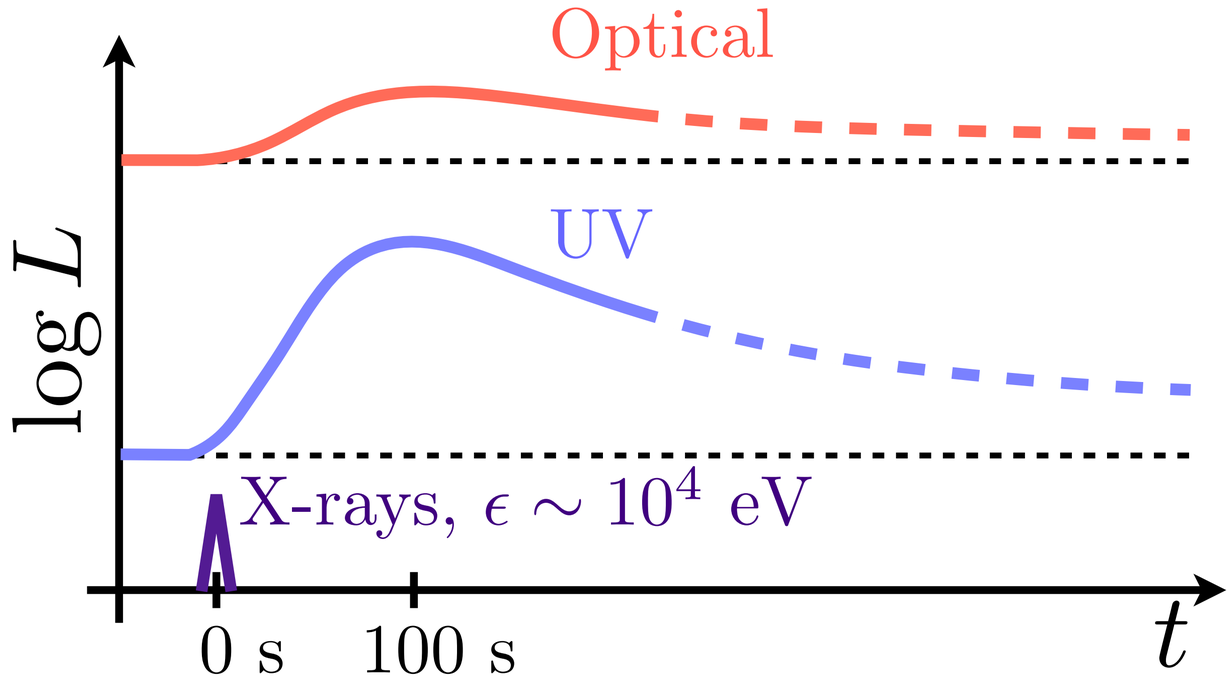


Typical shocked surface temperature

Energy **released** from star surface

Point of shock calculation: typical final temperature is UV; energy released is suppressed by shock dissipation as $R_{\rm DM}$ decreases, but still sizable

Detecting the Signal



Main signature: thermal UV transient spread over ~hundreds of seconds

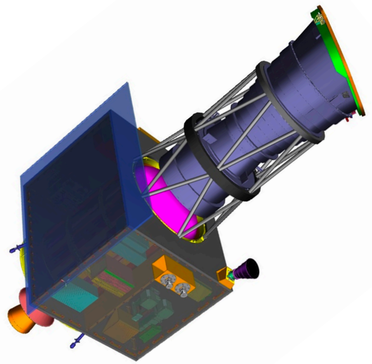
Search Strategies

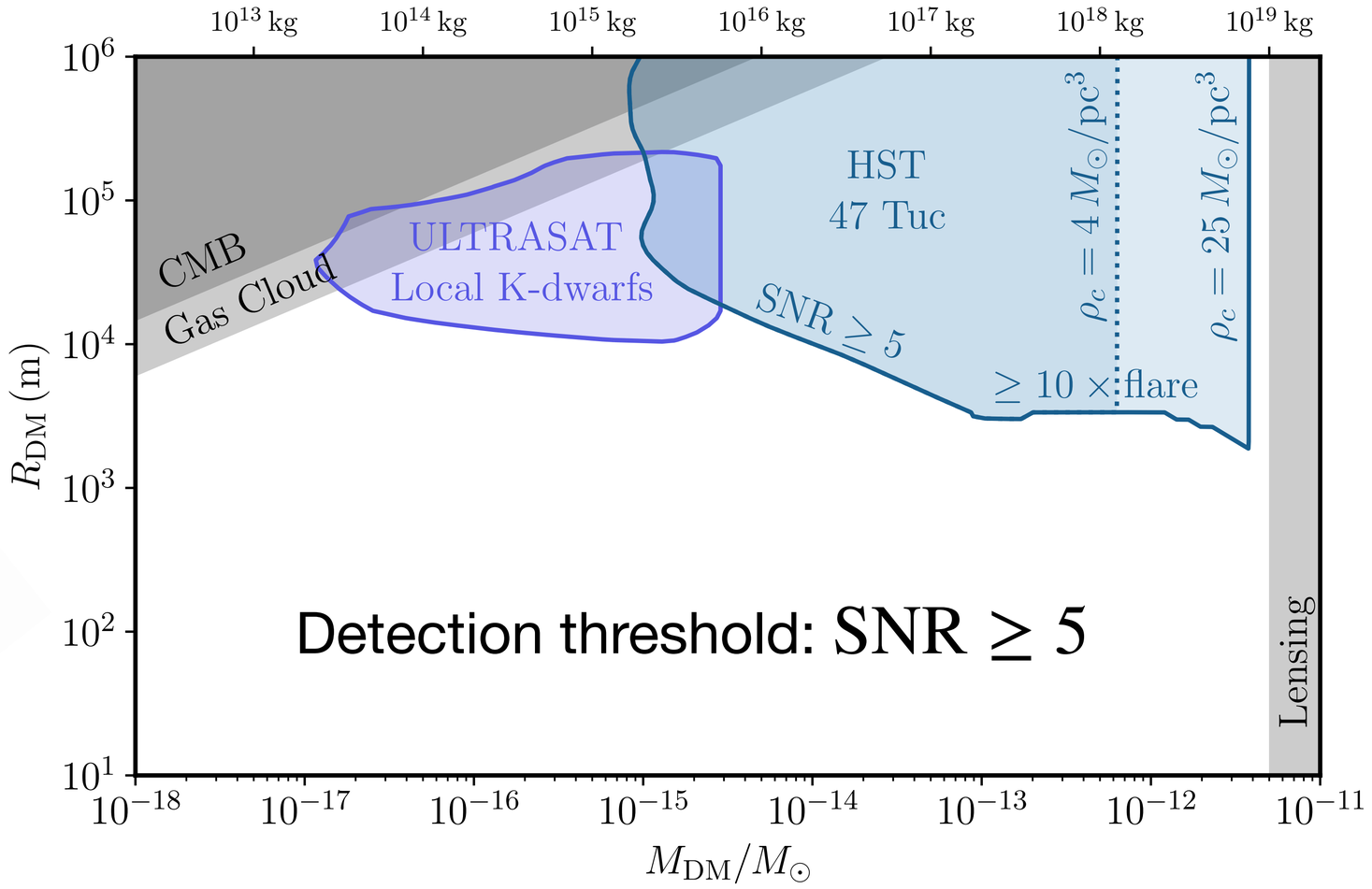
Local search

- Wide field search for transients on nearby (within $1\,\mathrm{kpc}$) stars
- Does not require dedicated survey
- Rate low, so targets low M_{DM}
- Stellar superflare background could be high

Focused search

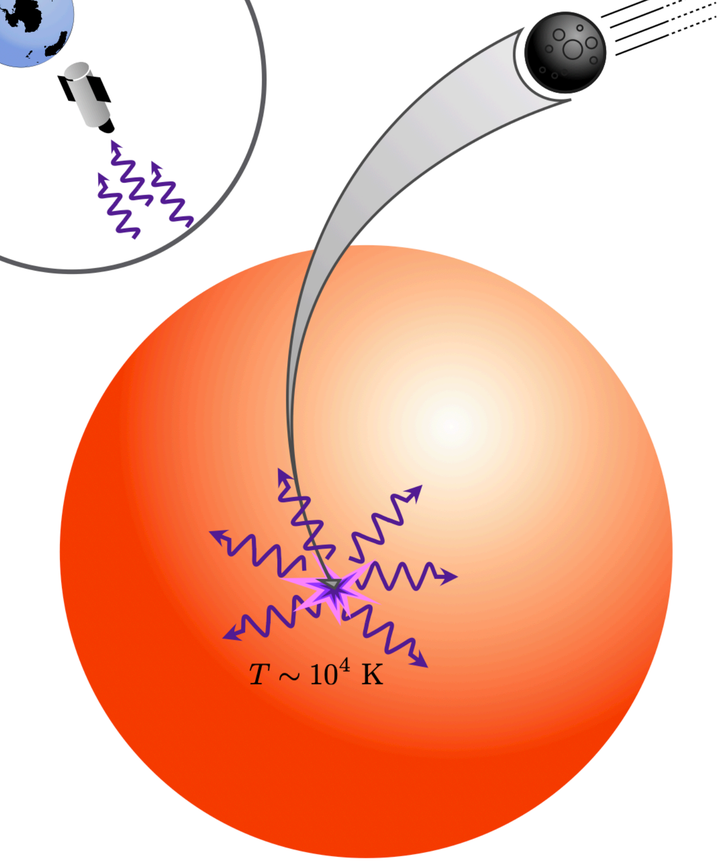
- Point to region with high $\Gamma \propto \rho_{\mathrm{DM}}/v_{\mathrm{DM}}$
- Requires dedicated search with powerful UV telescope
- Multiple kpc away, so targets high M_{DM}
- Events very energetic, background negligible





Local search: planned UV transient telescope ULTRASAT, nearby K dwarfs

Focused search: Hubble Space Telescope, core of globular cluster 47 Tuc



Future Directions

- Other star types (brown dwarf) and telescopes (LSST)
- Constraints with archival data
- Other focused searches (galactic center, Milky Way satellites, other globular clusters)
- Hydrodynamic simulation of shock propagation and subsequent cooling

Popular Interest

About this Attention Score

In the top 5% of all research outputs scored by Altmetric

High Attention Score compared to outputs of the same age (98th percentile)

High Attention Score compared to outputs of the same age and source (97th percentile)

FEBRUARY 1, 2022 FEATURE

Dark matter travelling through stars could produce potentially detectable shock waves

"Dark matter asteroids" could be spotted colliding with stars

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FEBRUARY 1, 2022 FEATURE

Dark matter travelling through stars could produce potentially detectable shock waves

"Dark matter asteroids" could be spotted colliding with stars

ASTROFÍSICA

La materia oscura podría generar ondas capaces de ser detectadas

Se identificarían mediante telescopios sofisticados

ASTEROIDEN AUS DUNKLER MATERIE (SOFERN VORHANDEN) KÖNNEN SONNENERUPTIONEN VERURSACHEN

LENTA.RU

Раскрыты последствия столкновения темной материи со звездой

Popular Responses

Rotogizmag FEBRUARY 2, 2022 11:29 AM

Personally I would not be surprised if dark matter = small size black holes including primordial formed during the bang. As confusions said "The hardest thing to do is to find a black cat in a dark room ... especially when there is no black cat.".

dollyllama / Ars Scholae Palatinae

Is dark matter God?

-17 (0 / -17)

Wizard of IT / Smack-Fu Master, in training

We all know Dark Matter is little balls that Nibblonians leave in the litter box.

+11 (+11 / 0)

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TheOldChevy / Ars Scholae Palatinae / *et Subscriptor*

If I look up, will I see the dark matter asteroid coming to destroy earth ?

+12 (+19 / -7)

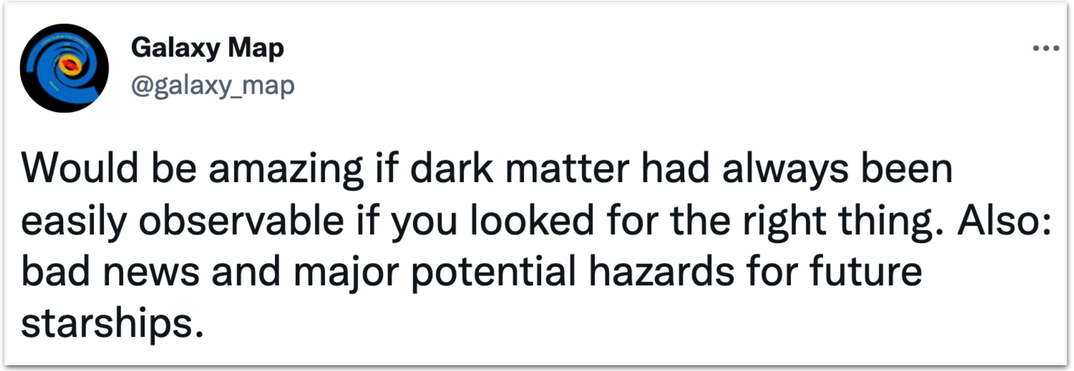
S_T_R / Ars Tribunus Militum

> TheOldChevy wrote:
> If I look up, will I see the dark matter asteroid coming to destroy earth ?

Only in theaters and even then only when Roland Emmerich gets around to it.

+61 (+62 / -1)

Popular Responses



Galaxy Map
@galaxy_map

Would be amazing if dark matter had always been easily observable if you looked for the right thing. Also: bad news and major potential hazards for future starships.

observicist 0

Feb 02, 2022

> [cont'd]
> Assuming such objects exist, and given the low local DM density, they would be rare in the vicinity of the solar system, so a search for possible effects of such objects colliding with other stars would seem plausible.

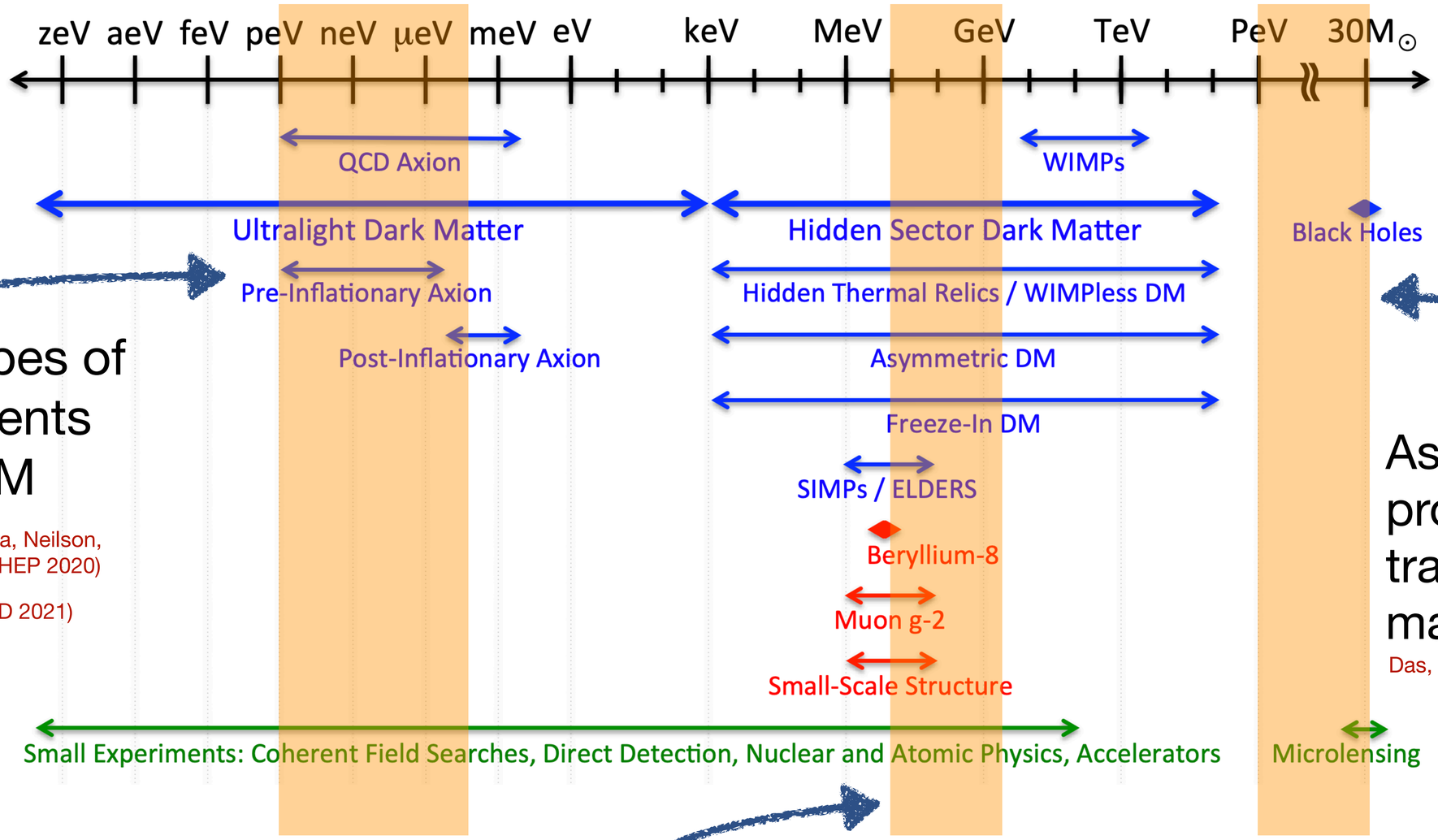
I entirely agree with both you and the article.

This stuff is hard to find, so we need to look for it in every way we can think of.

Something's there, and we need to find it.

Physics still captures the imagination!

The Dark Matter Landscape



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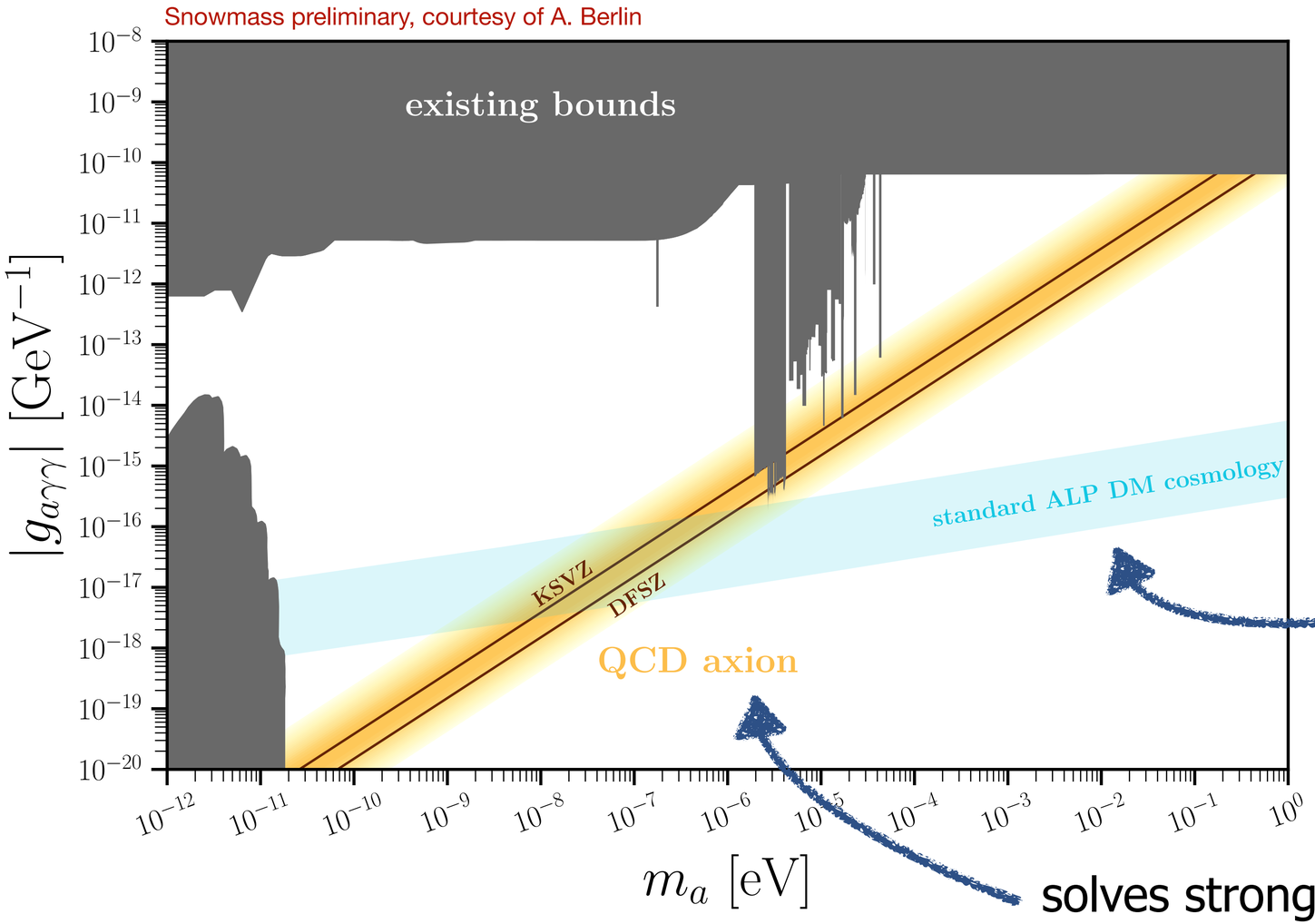
Vector meson decays to DM at LDMX

Schuster, Toro, KZ (PRD 2022)

Axion Parameters



Axions are light pseudoscalar fields protected by a shift symmetry

Theoretically well-motivated, with simple and predictive dark matter production

produced in right abundance with $\mathcal{O}(1)$ misalignment angle, temperature independent mass

solves strong CP problem (original motivation)

Electromagnetically Coupled Axions

Axions generically have coupling $\mathscr{L} \supset gaF\tilde{F}$, which modifies Ampere's law to

$$\nabla \times \mathbf{B} = \dot{\mathbf{E}} + \mathbf{J} + g\dot{a}\mathbf{B}$$

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$$\nabla \times \mathbf{B} = \dot{\mathbf{E}} + \mathbf{J} + g\dot{a}\mathbf{B}$$

Axion field is $a(t) \simeq a_0 \cos(m_a t)$, so in background field, creates effective current

$$J_{\text{eff}} \sim g m_a a B_0 \sim g\sqrt{\rho_{\text{dm}}}\, B_0$$

If background field occupies length R, this creates field $B_a \sim J_{\text{eff}} R$
(as long as no destructive interference, current frequency $\omega \lesssim 1/R$)

Electromagnetically Coupled Axions

$$J_{\text{eff}} \sim g\sqrt{\rho_{\text{dm}}}\, B_0 \qquad\qquad B_a \sim J_{\text{eff}} R$$

Most axion searches resonantly amplify this field by Q, giving

$$P_{\text{sig}} \sim \frac{\omega U}{Q} \sim \frac{\omega\,(QB_a)^2\, V}{Q}$$

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$$P_{\text{sig}} \sim \frac{\omega U}{Q} \sim \frac{\omega\,(QB_a)^2\,V}{Q} \sim \frac{g^2\rho_{\text{DM}}}{m_a}\,(QB_0^2V)\,(m_a\omega R^2)$$

Fixed properties of the DM

Properties of the resonator (scales with $)

Dimensionless factor, depends on experimental design

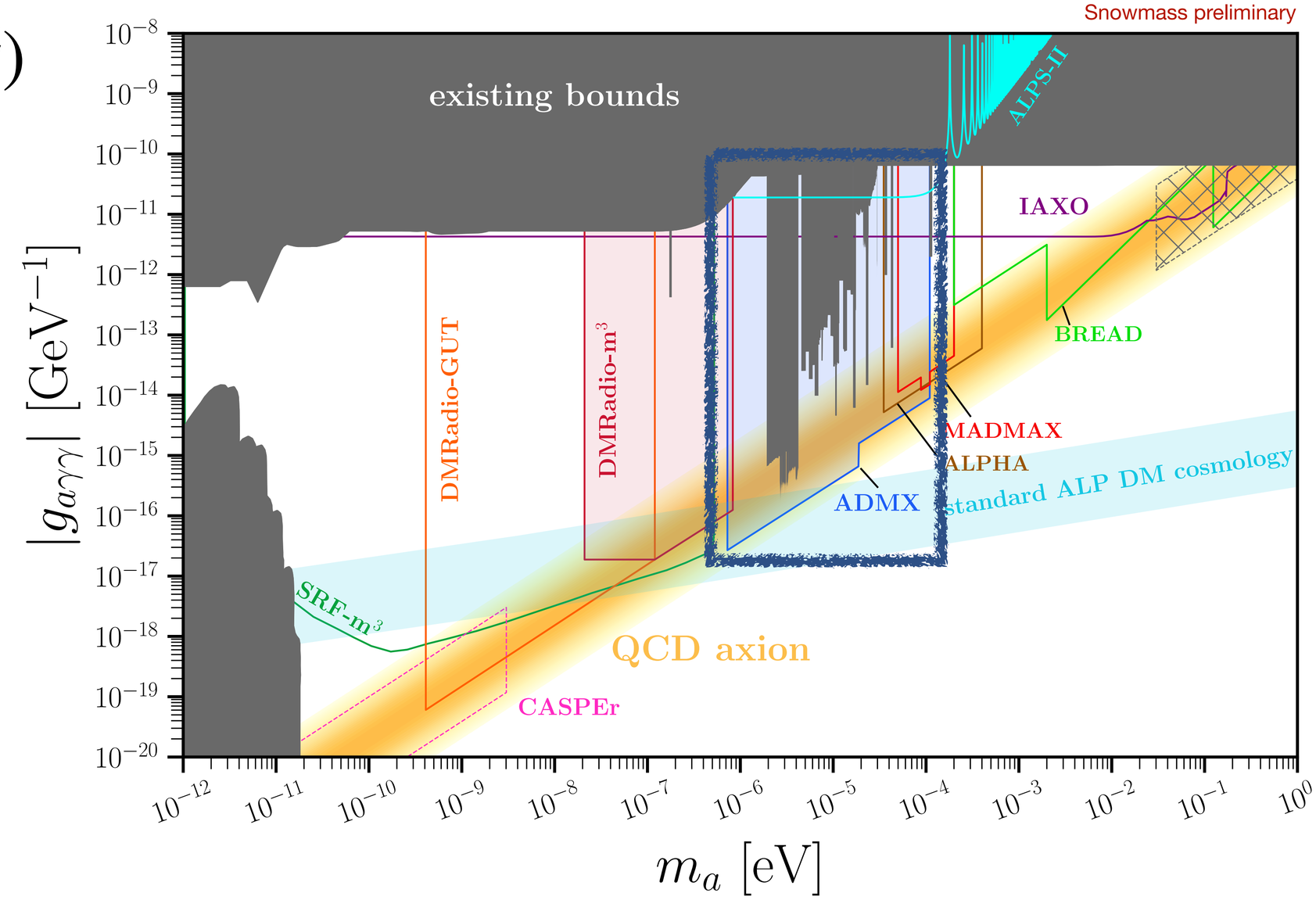
Three Resonator Designs

$$P_{\text{sig}} \propto (QB_0^2V)\,(m_a\omega R^2)$$

Traditional resonant cavity:

has $\omega = m_a$

works for $m_a R \sim 1$



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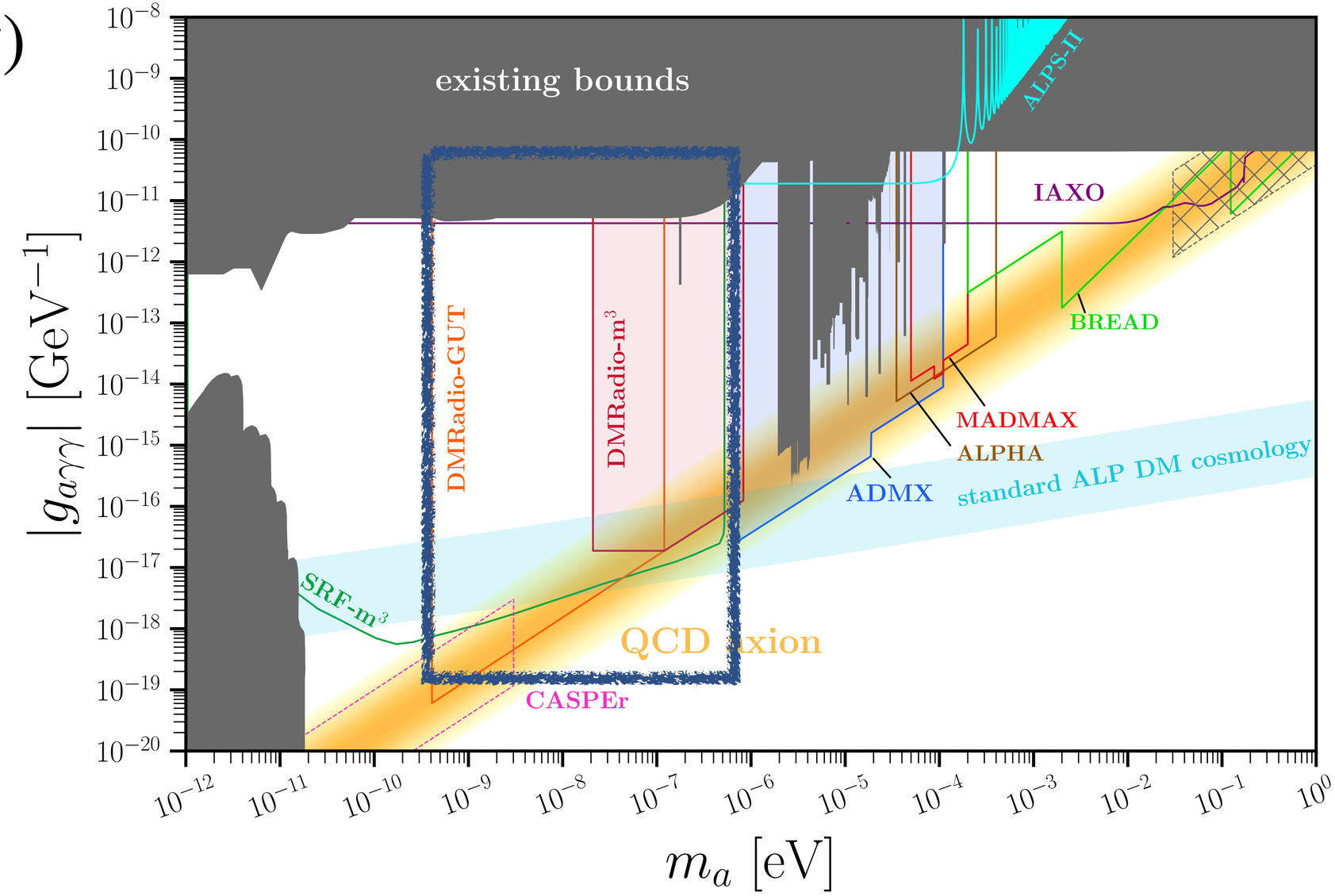

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has $\omega = m_a$

works for $m_a R \sim 1$

Quasistatic LC resonator:

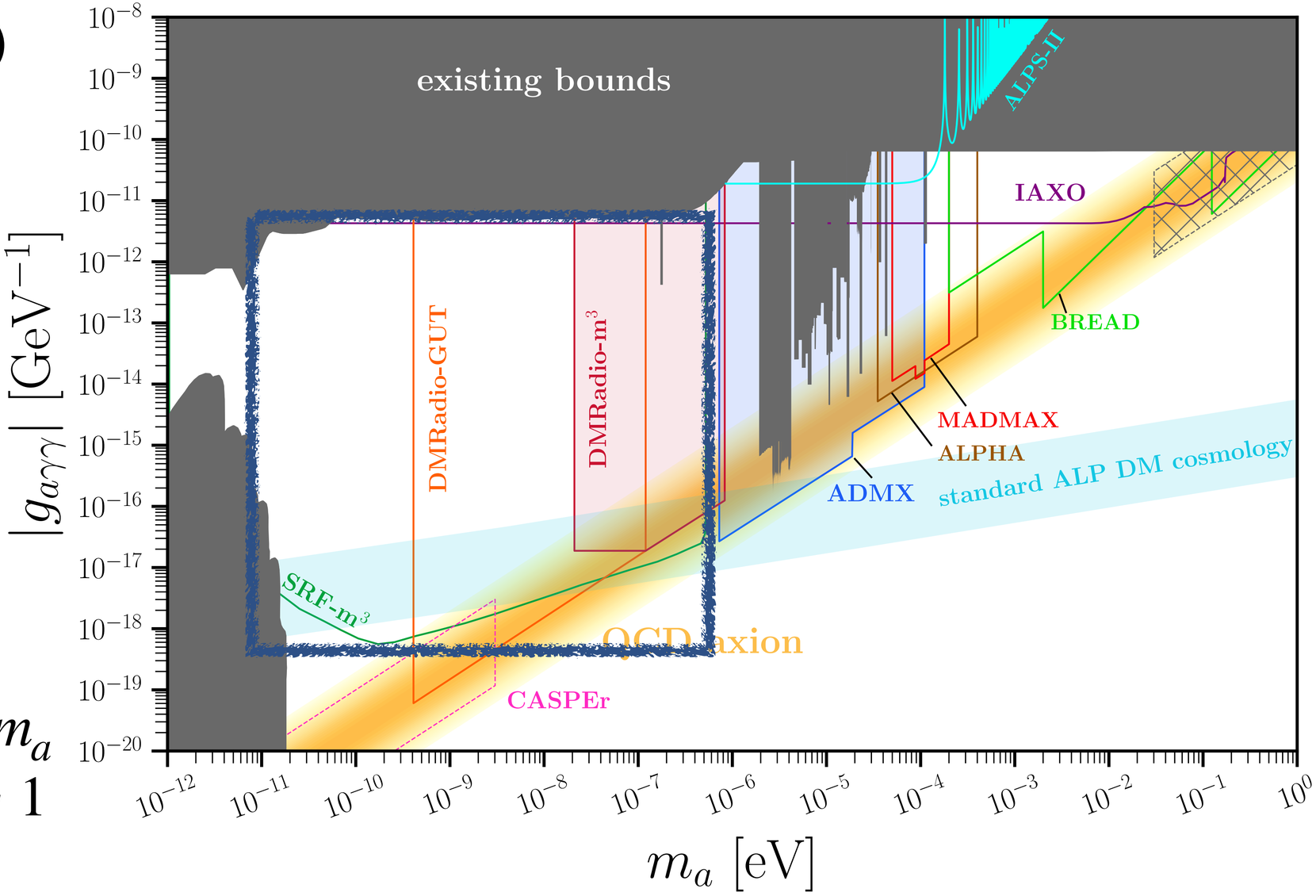
has $\omega = m_a$

works for $m_a R \ll 1$

Heterodyne approach:

oscillating B_0, $\omega \approx \omega_0 \gg m_a$

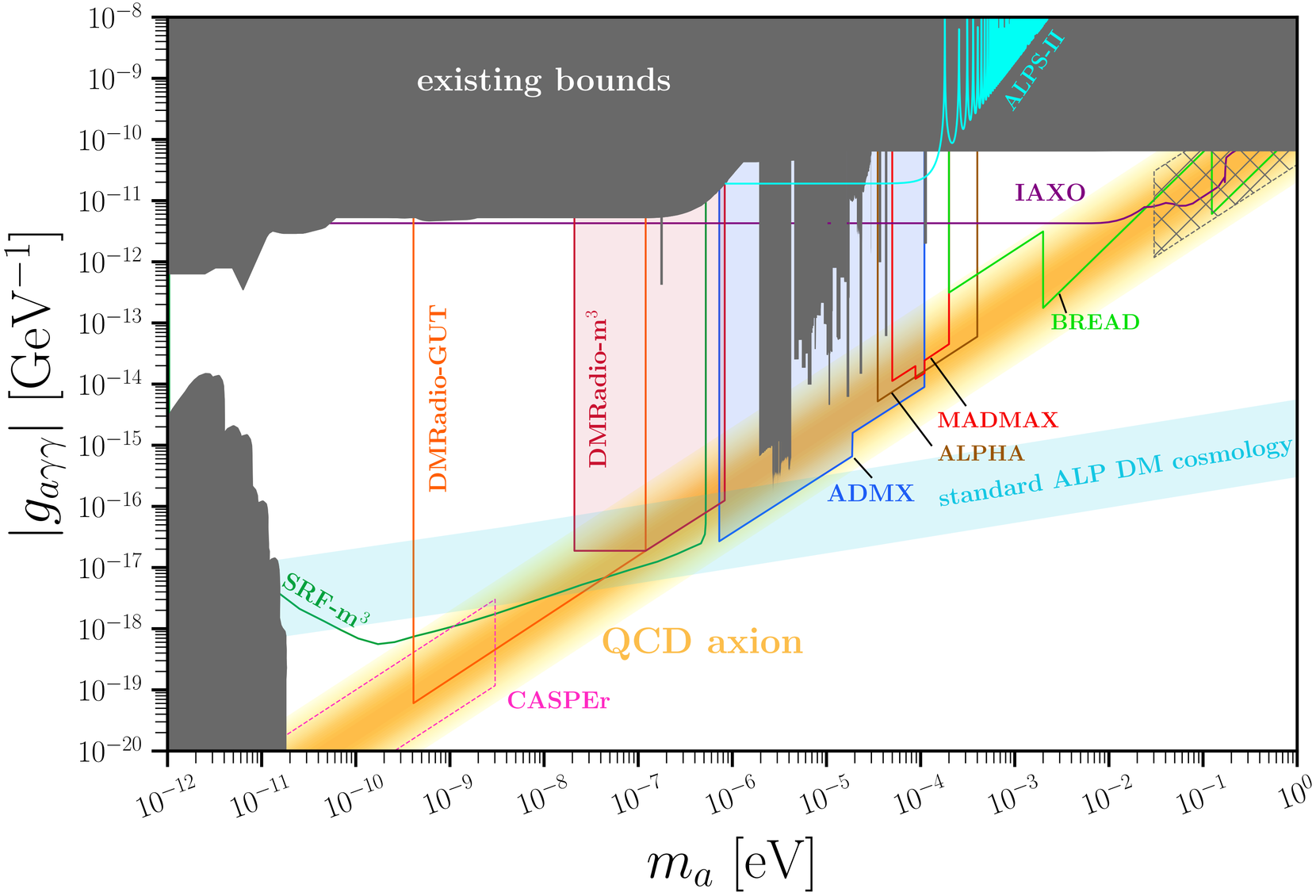
works for $m_a R \ll 1$, $\omega_0 R \sim 1$



Three Resonator Designs

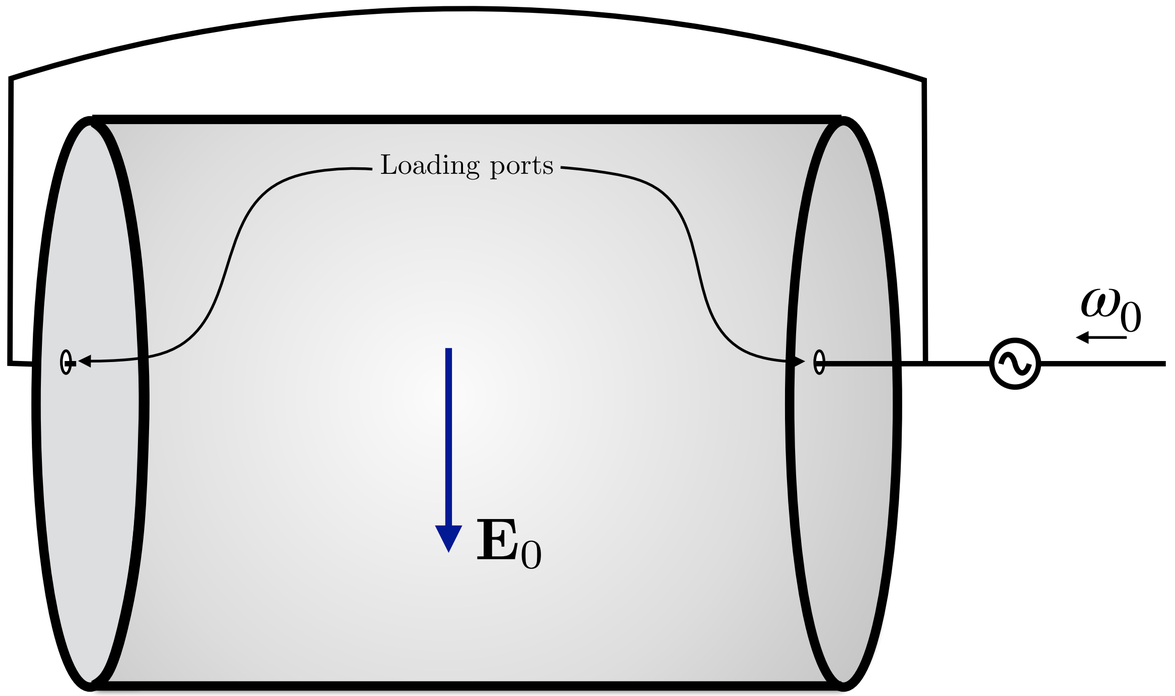
$$P_{\text{sig}} \propto (QB_0^2V) \times \begin{cases} 1 & \text{cavity} \\ (m_aR)^2 & \text{LC} \\ m_aR & \text{hetero} \end{cases}$$

Heterodyne approach can reach low m_a like LC circuit, but has less suppressed power, like traditional cavity!



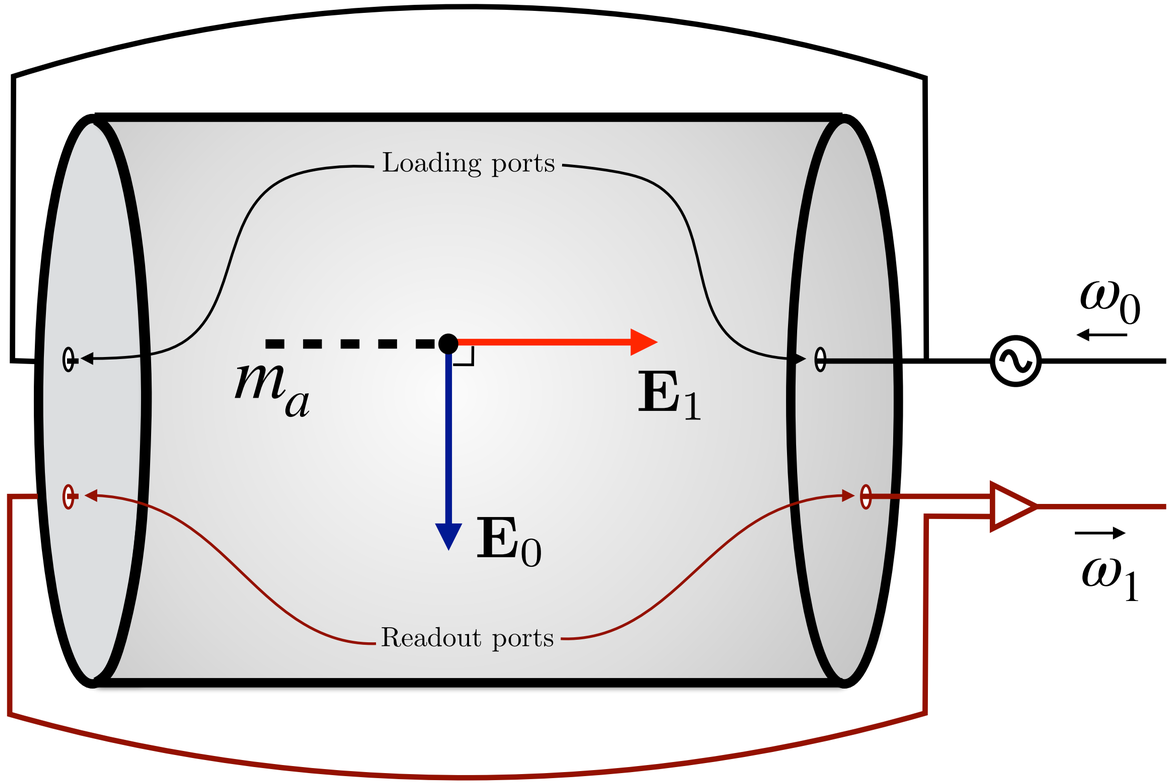
Implementing the Heterodyne Approach

Load a cavity mode with frequency $\omega_0 \sim \mathrm{GHz}$



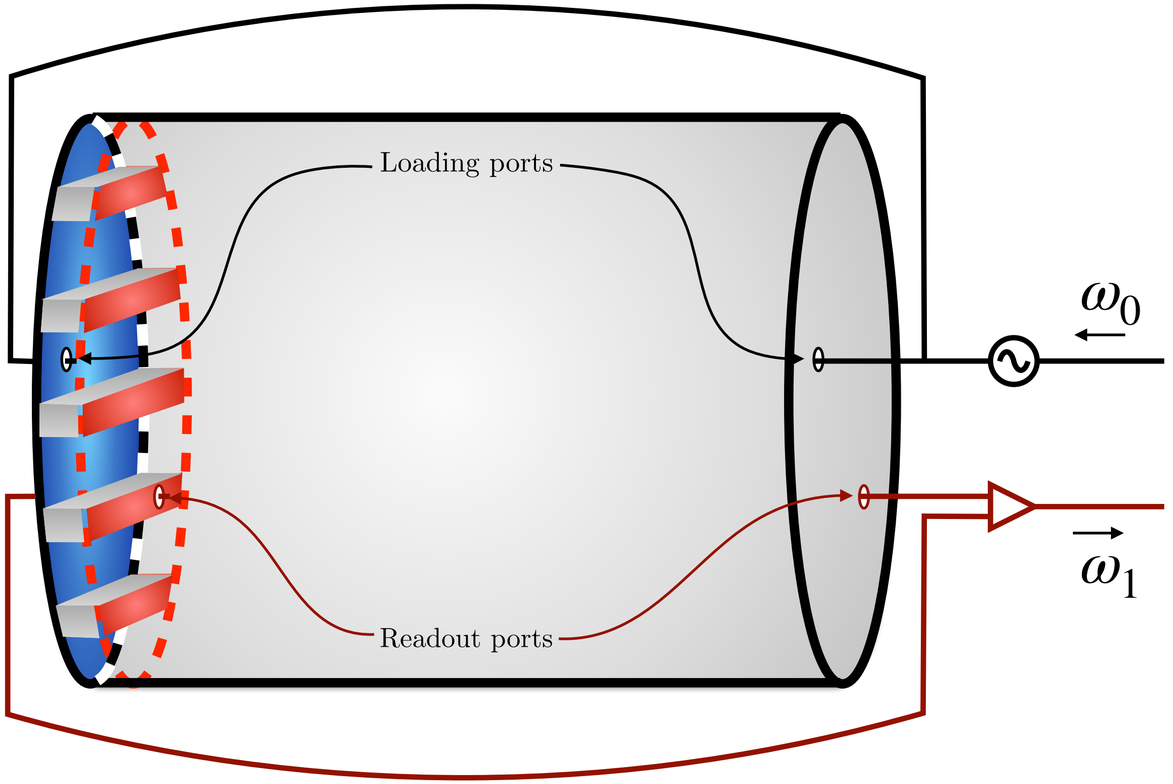
Implementing the Heterodyne Approach

Axion J_{eff} resonantly drives an orthogonal "signal" mode with frequency $\omega_1 = \omega_0 + m_a$



Implementing the Heterodyne Approach

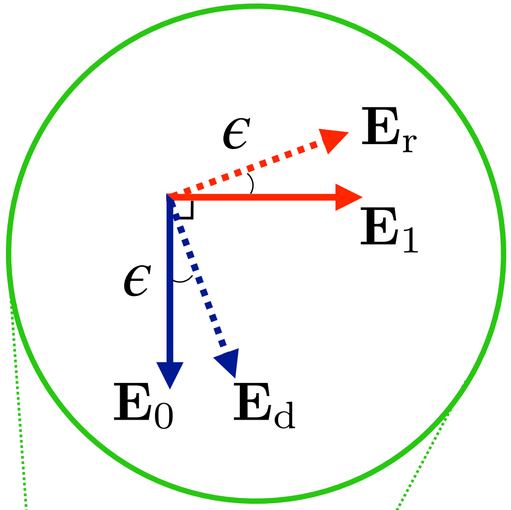
Adjust the frequency difference to scan in m_a by perturbing cavity walls



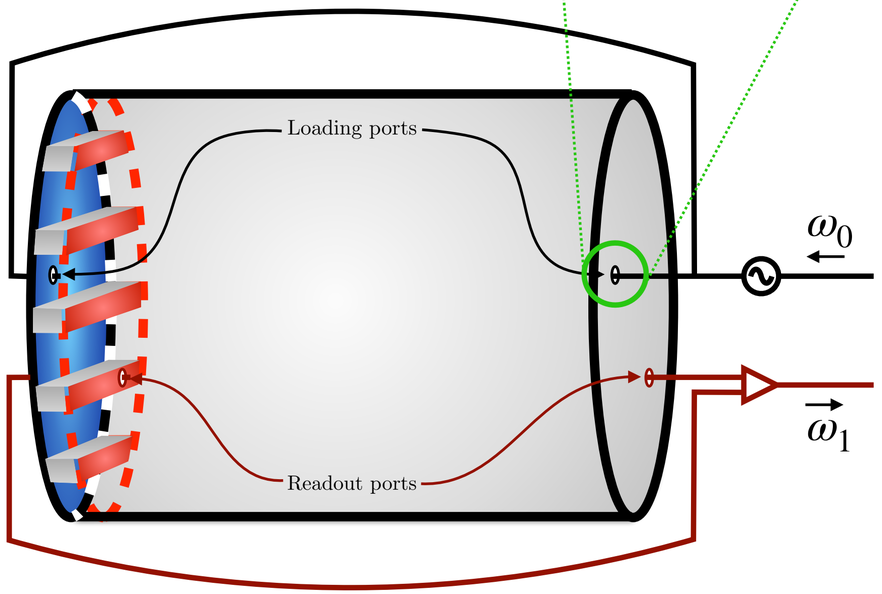
Main noise source at low m_a is leakage: environmental noise sources at frequency m_a can also be converted to frequency ω_1

Suppressed by small cross-coupling ϵ

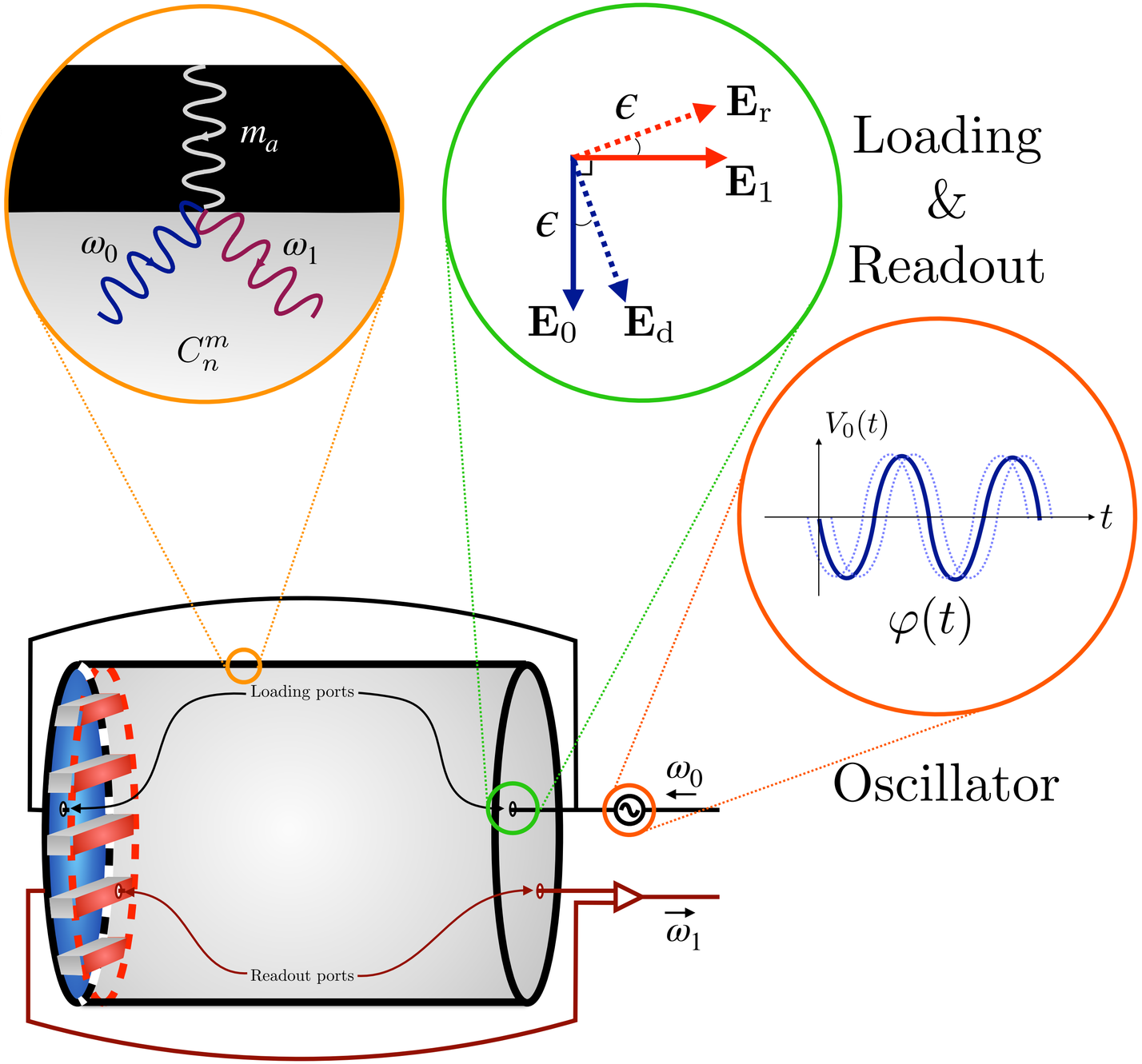
Use SRF cavity to lower input power ($Q \sim 10^{12}$)

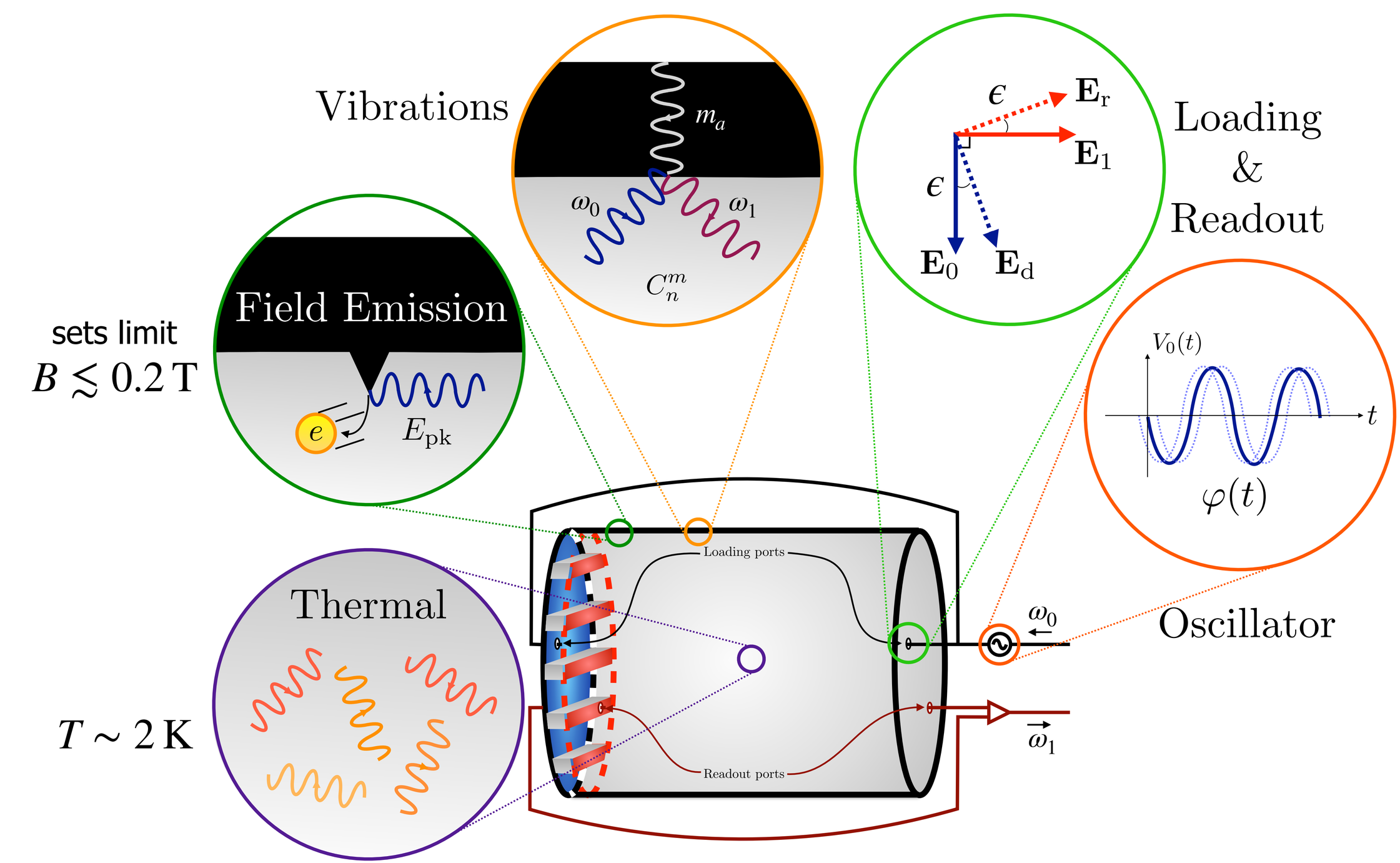


Loading & Readout

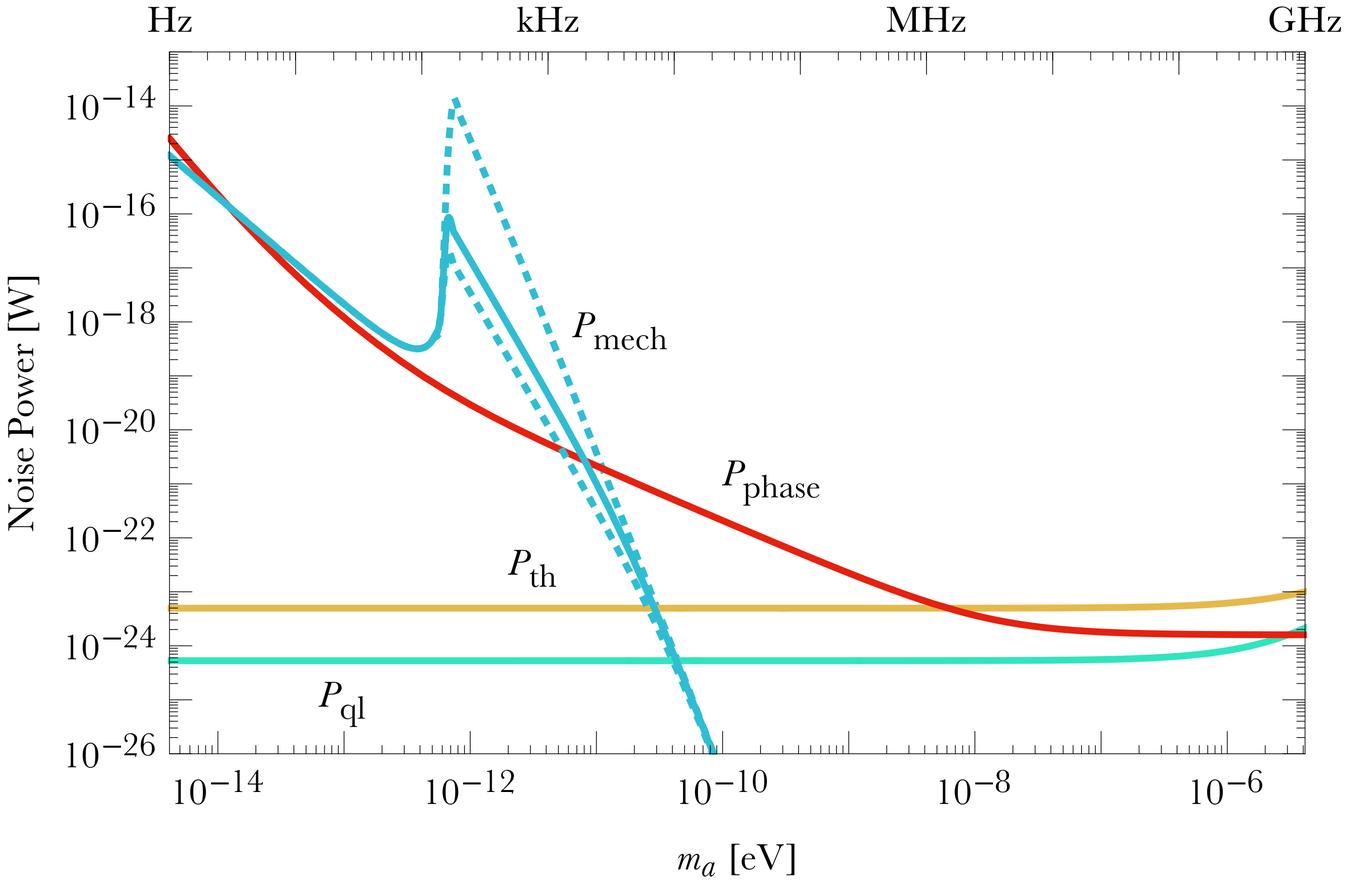


Leakage comes from wall vibrations (cooling system, piezo tuners), phase noise in oscillator driving the cavity




Vibrations
m_a
ω_0
ω_1
C_n^m
Loading & Readout
ϵ
$\mathbf{E}_\mathrm{r}$
$\mathbf{E}_1$
ϵ
$\mathbf{E}_0$
$\mathbf{E}_\mathrm{d}$
Field Emission
sets limit
$B \lesssim 0.2\,\mathrm{T}$
e
E_pk
$V_0(t)$
t
$\varphi(t)$
Loading ports
Readout ports
ω_0
$\vec{\omega}_1$
Oscillator
Thermal
$T \sim 2\,\mathrm{K}$

Noise Estimates



Wall vibrations dominate at low frequencies

Oscillator phase noise dominates at medium frequencies

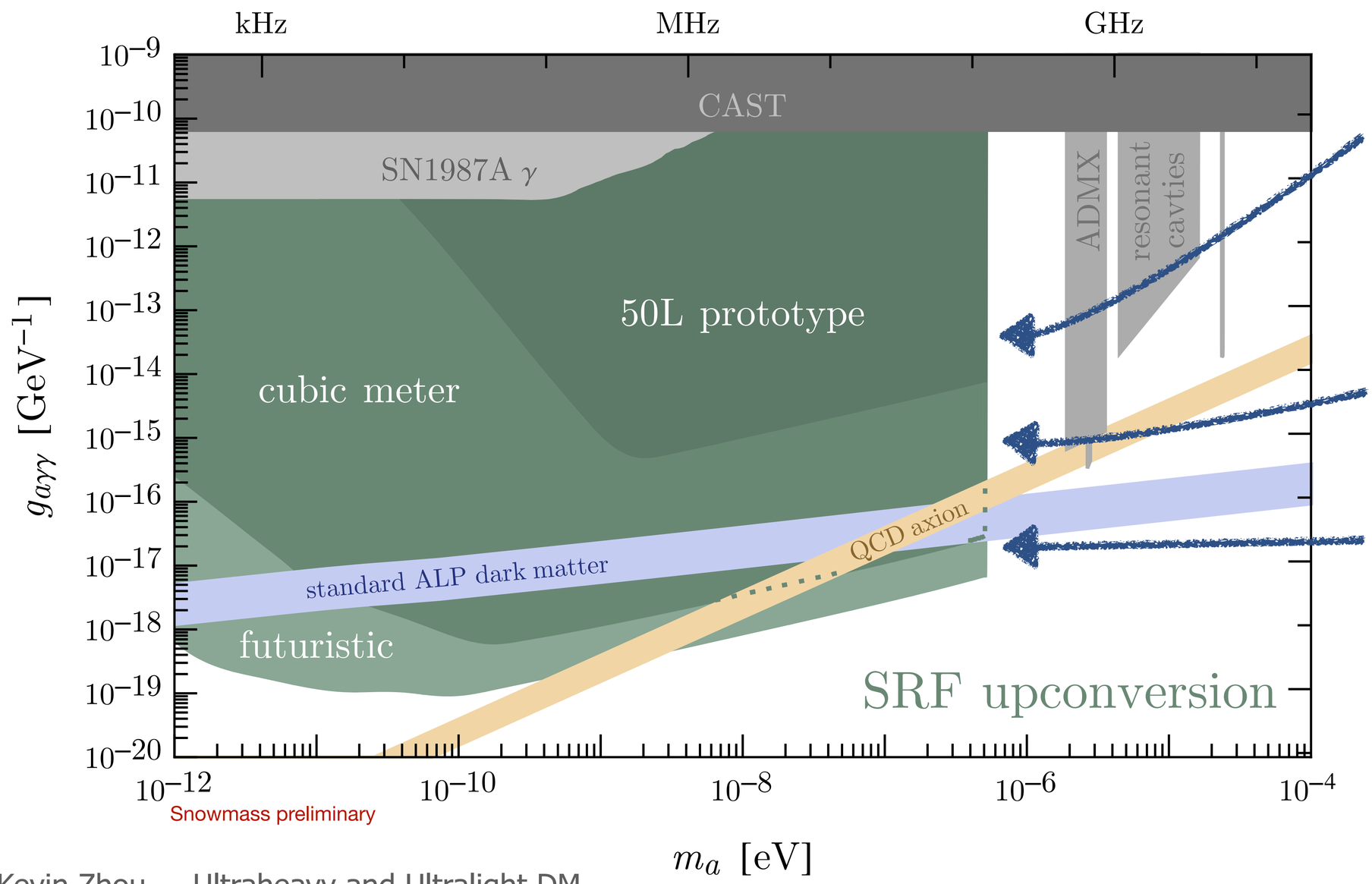
Thermal/amplifier noise dominates at high frequencies

Predecessors

MAGO (2005): heterodyne gravitational wave detection, reported $\epsilon \sim 10^{-7}$

DarkSRF (2020): dark photon light-shining-through-wall experiment with SRF cavities, stabilized to $\Delta\omega/\omega \sim 10^{-10}$

Projected Reach



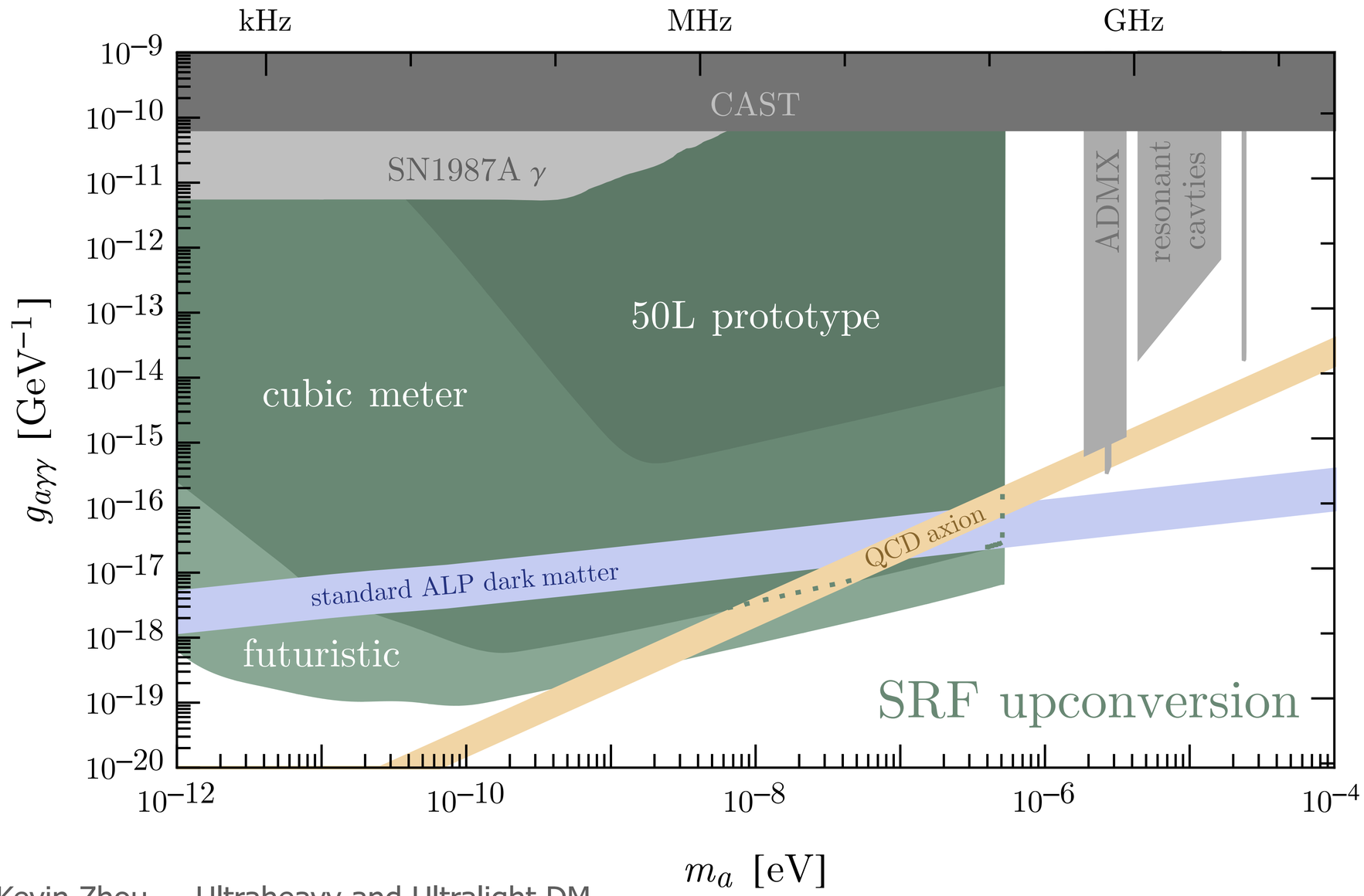
possible with “off the shelf” technology, existing cavities, no vibration attenuation

reach of dedicated program assuming best values each realized in other experiments

$\epsilon = 10^{-7}$, $Q = 10^{12}$

“best possible” future reach

Outlook

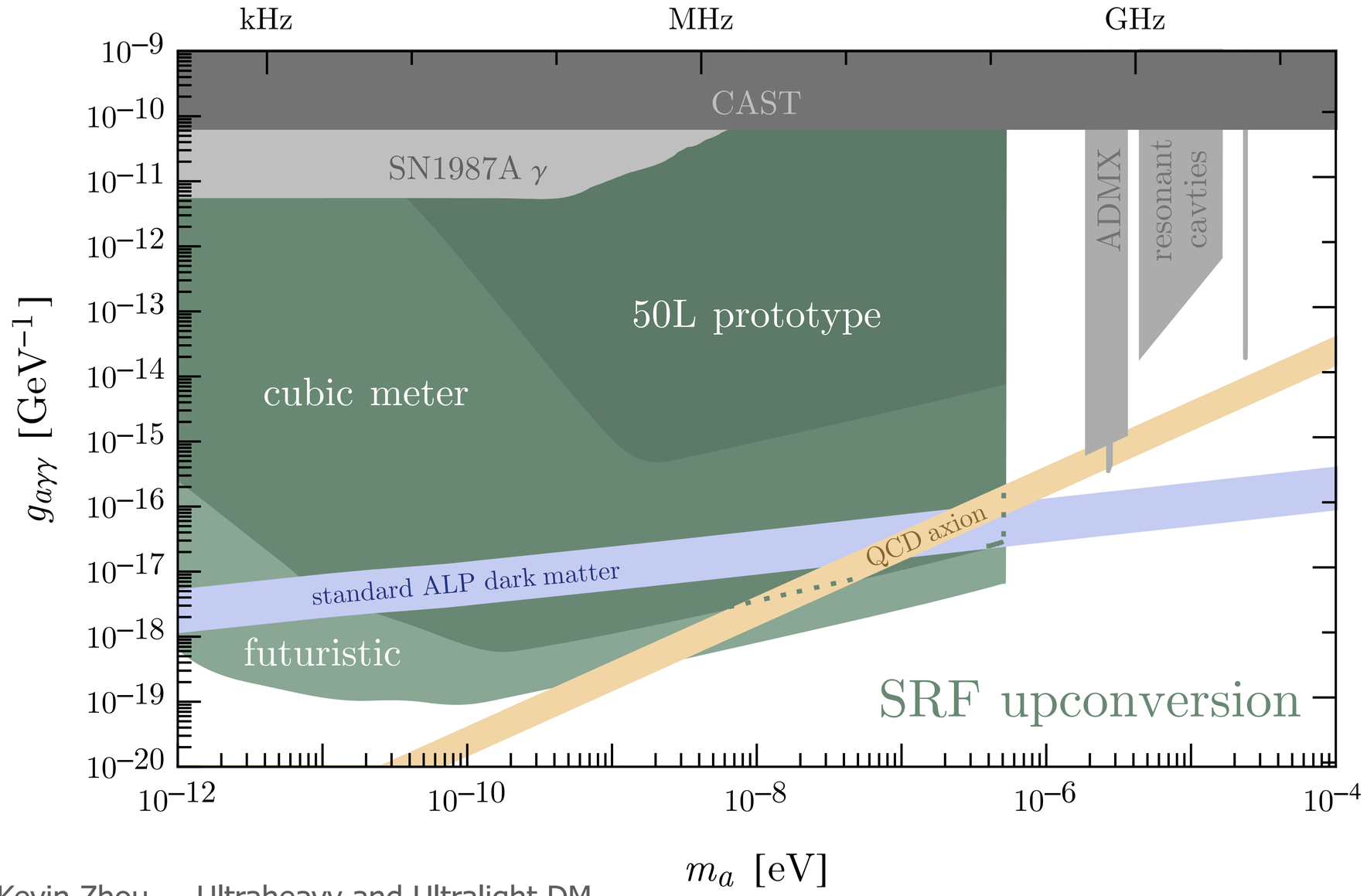


Heterodyne approach leverages SRF cavity advances

Higher signal power: don't need extreme cooling or quantum metrology

New noise sources: require careful study

Outlook



Cavity/tuner design study funded by LDRD grant at SLAC

Related prototype built at Australia's ARC DM Centre (PRL 126, 081803 (2021))

Interest in realistic measurements of leakage noise at Fermilab and CERN